

1993

American Petroleum Institute

RP 11S4

Recommended Practice for Sizing and Selection of Electric Submersible Pump Installations

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API RECOMMENDED PRACTICE 11S4 (RP 11S4)

FIRST EDITION, MAY 1, 1993

(Formally API Recommended Practice 11U (RP 11U)

Second Edition

Reaffirmed and Reassigned, May 1, 1993)

American Petroleum Institute

1220 L Street, Northwest

Washington, DC 20005

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Issued by

AMERICAN PETROLEUM INSTITUTE

Production Department

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Note: This is the first edition of this API Recommended Practice. It supersedes API Recommended Practice 11U (RP 11U) second edition, May 30, 1986 which was submitted to be renamed and reaffirmed at the 1992 Standardization Conference as reported in Circ PS-1985, and subsequently approved by letter ballot.

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Foreword

- 1. Jurisdiction:** This Recommended Practice (RP) is under the jurisdiction of the American Petroleum Institute (API) Committee on Standardization of Production Equipment (Committee 11).
- 2. Purpose:** This document presents Recommended Practices for Sizing and Selection of Electric Submersible Pump Installations and Related Equipment.

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SECTION 1 GENERAL

1.1 INTRODUCTION:

1.1.1

This RP is provided to meet the need for guidelines, procedures and recommendations covering Electric Submersible Pumps and related equipment. These Recommended Practices are those generally considered necessary for successful electric submersible pump operation. For installation see API RP 11R, *Recommended Practice for Electric Submersible Pump Installations*.

1.2 SCOPE:

1.2.1

This Recommended Practice addresses the fundamental procedures for sizing and selecting electric submersible pumps and related equipment for various well conditions.

1.2.2

This Recommended Practice is specifically prepared for installations in oil and/or water producing wells where the equipment is installed on tubing.

SECTION 2 TERMS, DATA AND CHARTS

2.1 DEFINITION OF TERMS:

2.1.1 VD — Vertical Pump Intake Setting Depth (Ft.)

— is the true vertical pump setting depth measured from the surface.

2.1.2 MD — Measured Pump Intake Setting Depth (Ft.)

— actual pump setting depth measured along the well bore. Theoretically, in non-deviated wells, MD = VD.

2.1.3 FOP — Fluid Over Pump (Ft.)

— calculated value

$$\text{FOP} = \frac{\text{PIP (PSI)}}{\text{S.G.} \times 0.433 \text{ (PSI/Ft.)}}$$

where: PIP = Pump Intake Pressure (PSI)
S.G. = Pump Intake Fluid Specific Gravity
0.433 = Conversion Constant (PSI/Ft.)

CAUTION: FOP calculated from a Sonic Fluid Level (FL) where $\text{FOP} = \text{VD} - \text{FL}$, can be very misleading because of the actual fluid gradient (PSI/Ft.) of the fluid in the annulus. This Fluid Level (FOP) may not give an equivalent Pump Intake Pressure.

2.1.4 H_D — Vertical Lift (Ft.)

— Vertical Pump Setting Depth (VD) minus Fluid Over Pump (FOP). Therefore:

$$H_D = \text{VD} - \text{FOP}$$

2.1.5 H_F — Tubing Friction Loss (Ft.)

— Pressure Loss expressed in feet due to fluid flow through the tubing, and is a function of fluid velocity

and viscosity.

2.1.6 H_T — Producing Tubing Head (Ft.)

— Producing tubing pressure at wellhead converted to head in feet by using the specific gravity of the fluid being pumped.

$$H_T = \frac{\text{Tbg. Press. (PSIG)}}{\text{S.G.} \times 0.433}$$

2.1.7 TDH — Total Dynamic Head (Ft.)

— is the head against which the pump is operating (discharging) expressed as feet of fluid being lifted.

$$\text{TDH} = H_D + H_F + H_T$$

where:

H_D = VD - FOP = Vertical Lift (Ft.)

H_F = Tubing Friction Loss (Ft.)

H_T = Producing Tubing Head (Ft.)

2.1.8 S.G. — Specific Gravity

— of oil, water and solution gas or any mixture thereof, at the pump intake.

2.1.9 Visc

— Viscosity (SSU)

2.1.10 BHT

— Bottom Hole Temperature (°F)

2.1.11 GOR

— Gas Oil Ratio (cu. ft./BBL)

2.1.12 STBO

— Stock Tank Barrels Oil (BBL)

2.1.13 CUT

— Water Cut (%)

2.1.14 PVF

— Pump Volume Factor

2.1.15 GLR

— Gas Liquid Ratio (cu. ft./BBL)

2.1.16 KVA

$$\text{KVA} = \frac{\text{Transformer Power Rating} \times 1.732 \times \text{Volts (Line-To-Line)} \times \text{Amps}}{1000}$$

2.1.17 R_s

— Volume of gas in solution at any pressure up to bubble point pressure (SCF/BBL). Bubble point pressure is defined as the pressure above which all gas is in solution.

$$R_s = Y_g \left[\frac{P}{18} \left(\frac{10^{0.0125 (\text{API}^\circ)}}{10^{0.00091t}} \right) \right]^{\frac{1}{0.83}}$$

Where: Y = Sp. Gr. of Gas
P = Bubble Point Pressure (PSI)
API = API Gravity of Oil ($^\circ\text{API}$)
t = Fluid Temperature ($^\circ\text{F}$)

2.1.18 B_o

— Shrinkage Factor (formation volume factor). A multiplier applied to oil stock tank barrels to determine volume occupied in formation. Can be calculated.

$$B_o = 0.972 + [0.000147F]^{1.175}$$

Where:

$$F = R_s \left[\frac{(Y_g)}{Y_o} \right]^{0.5} + 1.25t$$

Y_o = Sp. Gr. of Oil.

2.1.19 B_g or Formation Gas Volume Factor

— A multiplier applied to the volume of free gas existing at a given condition to determine the volume occupied in the formation of free gas.

$$B_g = .0283 \left[\frac{ZT}{P} \right] \text{ Ft.}^3/\text{SCF} \text{ or } .0283 \times \left[\frac{1728 \times 1000}{42 \times 231} \frac{ZT}{P} \right] \text{ bbls/MSCF}$$

$$B_g = 5.04 \frac{ZT}{P} \text{ BBLs/mcf}$$

Where: Z = Gas Compressibility Factor
T = Temperature °R
P = Pressure PSIA

2.1.20

Vo = Volume of oil at PIP bbls.

2.1.21

Vg = Volume of free gas at PIP, bbls.

2.1.22

Vw = Volume of water at PIP, bbls.

2.1.23

Vt = Total volume of mixture (Vo + Vg + Vw) at PIP, bbls.

2.1.24

Vg/Vt — Gas volume to total volume ratio.

2.2 DATA AND CHARTS

2.2.1 Pump Performance Curves

(Fig. 1):

- Show the Discharge Head, Brake Horsepower and Efficiency as a function of Flow Rate for one stage of each pump type. Some curves may be set for 100 stages.
- Are based on fresh water (S.G. = 1.0).
- Give a Recommended Capacity Range for pump operation.

- d. Are commonly available for both 50 Hertz (Hz) and 60 Hz motor operation.

2.2.2 Fluid Friction Chart

Fig. 2):

- a. Shows fluid friction loss as a function of tubing size and flow rate.
- b. Is shown for water for a pipe roughness of $C = 120$. If a viscous fluid is involved, proper corrections must be made (see Section 2.2.3).

NOTE: A pipe roughness of $C = 120$ is used due to data availability and its universal use with non-viscous fluids.

2.2.3 Tubing Friction & Viscosity Correction Charts

(Figs. 3 - 10):

- a. It is recommended that viscosity correction be considered for pumped fluids with viscosity greater than 50 SSU.
- b. Fig. 3 may be used to determine viscosity (in centipoise) from gas free crude API gravity.
- c. Figure 4 may be used to determine gas solution for a given API gravity crude oil at pump intake pressure.
- d. Figure 5 using gas in solution allows the determination of the correction crude oil viscosity centipoise.
- e. Figure 6 may be used to convert viscosity from centipoise to SSU.
- f. Figure 7 provides the viscosity correction factor considering percent water cut and type of emulsion. Where water is the primary phase it wets the wall of the tubing and the viscosity will be dependent principally on the water properties.
- g. Figures 8, 9 and 10 give the viscous flow head loss due to fluid friction in various tubing sizes, fluid flow rates and fluid viscosities. (Pipe roughness $C = 130$).

NOTE: A pipe roughness of $C = 130$ is used due to data availability. Variance from a $C = 120$ is insignificant.

2.2.4 Pump Performance Correction Chart

(Fig. 11):

- a. It is recommended that viscosity correction should be considered to correct Pump Performance

when the viscosity of pumped fluids is greater than 50 SSU.

- b. Chart shown in Fig. 11 is to be used as general guideline only, since these values may change due to specific pump design and pump size. Equipment manufacturer should be consulted for proper correction factor.

2.2.5 Voltage Loss Chart

(Fig. 12):

- a. This chart gives the voltage loss per 1000 ft. for various sizes of power cable at 68°F. A correction factor should be used to determine the additional voltage loss at well operating temperature.

2.2.6 Well Temperature vs. Current Charts

(Figs. 13 - 16):

These charts are used to determine insulation temperatures when considering well temperature at bottom and current through cable.

2.2.7 Pump Volume Factor

(Figs. 17 - 18):

- a. These graphs allow calculation of the Pump Volume Factor (PVF) of a bubble point liquid knowing amount of gas in solution at Pump Intake Pressure (PIP) (Fig. 4), gas gravity, API gravity of the crude oil, and Bottom Hole Temperature (BHT).
- b. These graphs assume that all free gas is produced through the casing annulus.

2.2.8 Motor Annular Fluid Velocity

(Fig. 19):

- a. This chart allows the determination of fluid flow velocity by the motor for various casing sizes.

SECTION 3 ELECTRIC SUBMERSIBLE PUMP COMPONENT DESCRIPTION

3.1 Subsurface Equipment

(Fig. 20):

3.1.1 Motor

- a. The motor is the driving force (prime mover) which turns the pump, and is a two pole, three-phase, squirrel cage induction type that runs at a relatively constant speed of 3500 RPM at 60 Hertz

frequency power.

- b.** The motor is filled with a highly refined mineral oil or synthetic oil that must provide dielectric strength, lubrication for bearings, and good thermal conductivity. The nonconductive oil in the motor housing lubricates the motor bearings and transfers heat generated in the motor to the motor housing.
- c.** Heat from the motor housing is dissipated by the well fluids moving past the exterior surface of the motor. Therefore, a pumping unit motor should never be set below the point of fluid entry unless some means of directing the fluid by the motor is utilized. It is generally accepted practice to maintain a minimum fluid velocity of one ft/second to insure proper motor cooling.
- d.** Tandem motors are single motors electrically and mechanically connected in series to provide larger horsepower.
- e.** The motor is fitted with a thrust bearing which makes the motor suitable for rotation in one direction only as designed by the manufacturer.

3.1.2 Motor Seal Chamber

The motor seal chamber (or motor protector) in general performs the following five basic functions:

- a.** To connect pump housing to the motor housing and to connect the drive shaft of the motor to the pump shaft.
- b.** To house the pump thrust bearing to carry the axial thrust developed by the pump.
- c.** To prevent the entry of well fluid into the motor.
- d.** To equalize the pressure inside the motor with the well bore pressure and thus to eliminate the pressure difference across shaft seals.
- e.** To compensate for the expansion and contraction of motor oil due to heating and cooling of motor when the unit is running or shut down.

3.1.3 Intake Section or Gas Separator

The intake section plays the role of a suction manifold feeding the well fluid to the pump. Depending on well conditions, it can be in the form of a simple adapter with inlet holes or in the form of a gas separator. The gas separator is an aid in preventing gas lock and provides more efficient pumping of gassy wells. Their effectiveness varies with each design and produced volumes.

3.1.4 Pump

- a. The submersible pump is a multi-staged centrifugal type. Each stage consists of a rotating impeller and a stationary diffuser. The type and size of stage used determines the volume of fluid to be produced. The number of stages determines the total head generated and when multiplied by the horsepower per stage and specific gravity, the motor horsepower required can be determined.
- b. Pumps are manufactured in a broad range of capacities.
- c. Tandem pumps are two or more single pumps which are mechanically and hydraulically connected in series to provide the total pump discharge head required for the application.

3.1.5 Motor Lead Extension

The motor lead extension is a special low profile (flat) power cable extending from the pothead on the motor to above the end of the pump where it connects with the power cable. A low profile cable is usually needed in this area due to limited clearance between the pump housing and the well casing.

3.1.6 Motor Lead Extension Guards

The guards provide mechanical protection for the motor lead extension.

3.1.7 Power Cable

The power cable is a special three conductor cable which extends from the top of the pump to the wellhead, held to tubing by cable bands. It is available in a range of conductor sizes, insulation types and construction to accommodate well conditions and temperatures. Economic consideration governed by the cost of power and initial cost of cable must be evaluated.

3.1.8 Check Valves and Bleeder or Drain Valves

- a. If a check valve is installed, it should be located a minimum of 6 to 8 joints above the pump to allow gas to be purged from pump on start-up.
- b. If a check valve is installed, a bleeder or drain valve should also be installed. It should be located one joint above the check valve.
- c. For larger horsepower units (above 250 HP) regardless of setting depth, and low capacity units (less than 600 B/D) set deep (with static fluid level below 7000 feet), it is recommended that the tubing be filled before start-up. This means these installations must be equipped with tubing Check Valves and drain valves.

3.1.9 Cable Bands

- a. Cable bands are used to secure the Motor Lead Cable and Power Cable to the pump and tubing, respectively.
- b. Bands are available in carbon steel, stainless steel, and monel.
- c. Bands are available in 22" to 42" lengths or as continuous banding material.

3.2 SURFACE EQUIPMENT

(Fig. 21):

3.2.1 Wellhead

- a. The wellhead assembly provides a tubing support and allows the power cable to pass from the surface into the well bore.
- b. It must be provided with a wellhead feed through mandrel or pack-off method which provides a positive seal around the cable.
- c. Wellheads and cable pack-offs are available in various pressure ratings and hanging load capacities. This must be considered when selection is made, including any local safety codes or other requirements.
- d. All valves exposed to pump pressure must be rated at or above full pump discharge pressure.

3.2.2 Junction Box (Vent Box)

- a. A vented junction box provides a point to connect the surface cable from the motor control panel to the power cable from the well and provides a vent to the atmosphere for any gas which migrates through the power cable in oil wells.

3.2.3 Surface Cable

- a. Surface cable connects the junction box to the motor control panel and the motor control panel to the secondary side of the transformers.

3.2.4 Motor Control Panel

- a. A motor control panel is a combination motor starter, overload-underload protective device and recording instrument. Surge arrestors (Lightning Arrestors) should be included in the panel.
- b. They are available in various voltage and amperage ratings with additional controls according to power requirements.

3.2.5 Transformers

- a. The purpose of the transformer is to provide correct surface voltage for proper motor operation by changing electric power from a high voltage to a low voltage in the case of a step-down transformer or low to high in case of step-up transformer.
- b. Transformer selection is based on power rating in KVA, primary voltage, and tap arrangement.

SECTION 4 WELL DATA SHEET

(Fig. 22)

4.1

The proper selection of a particular pump type and size for a given application is very important and requires detailed information on well completion, reservoir data and production data and history.

4.1.1 Required Data

- a. The Well Data Sheet (Fig. 22) provides a form for collecting necessary data.
- b. This form should be filled out as completely as possible, making certain that all data provided is accurate, since this data will determine pump selection and sizing.
- c. Data lines preceded with an asterisk (*) indicate data that must be provided to allow pump sizing and selection.

SECTION 5 PUMP AND SYSTEM SIZING

5.1 Example Problem No. 1

— Water and Oil Production with no Gas. Problem nomenclature:

{definition term} $\frac{\text{Actual Value}}{\text{(reference number)}}$

5.1.1

The following exhibit sheets provide data and information required for Example Problem No. 1.

- a. Exhibit 1.1— Well Data Sheet
- b. Exhibit 1.2— Pump Performance Curve
- c. Exhibit 1.3— Pump Specifications

- d. Exhibit 1.4— Motor Specifications
- e. Exhibit 1.5— Motor Seal Chamber Specifications
- f. Exhibit 1.6— Motor Control Panel and Accessories Specifications
- g. Exhibit 1.7— Transformer Specifications
- h. Exhibit 1.8— Tubing Head Specification
- i. Exhibit 1.9— Cable Specifications
- j. Exhibit 1.10— Equipment and Cable Size Information
- k. Exhibit 1.11— Flat Cable and Surface Cable Specifications
- l. Exhibit 1.12— Downhole Equipment with Offset Tubing Head.

5.1.2

Unless otherwise specified, data used in sizing calculations comes from Well Data Sheet (Exhibit 1.1).

a. Well Fluid Composite Specific Gravity

$$\begin{aligned}
 1) & \left[\text{Water Cut} \right] \frac{0.80}{1} \times \left[\text{Water Sp. Gr.} \right] \frac{1.060}{(1)} = \frac{0.848}{(1)} \left[\begin{array}{c} \text{Water Phase} \\ \text{Sp. Gr.} \end{array} \right] \\
 2) & \left[\text{Oil Cut} \right] \frac{0.20}{1} \times \left[\text{Oil Sp. Gr.} \right] \frac{0.840}{(2)} = \frac{0.168}{(2)} \left[\begin{array}{c} \text{Oil Phase} \\ \text{Sp. Gr.} \end{array} \right] \\
 3) & \left[\text{Water Phase} \right] \frac{0.848}{(1)} + \left[\text{Oil Phase} \right] \frac{0.168}{(2)} = \frac{1.016}{(3)} \left[\begin{array}{c} \text{Well Fluid} \\ \text{Composite} \\ \text{Sp. Gr.} \end{array} \right]
 \end{aligned}$$

1) - 3)

b. Well Fluid Composite Gradient

$$1) \left[\begin{array}{c} \text{Well Fluid} \\ \text{Composite} \\ \text{Sp. Gr.} \end{array} \right] \frac{1.016}{(3)} \times \left[\begin{array}{c} \text{Fresh Water} \\ \text{Pressure} \\ \text{Gradient} \end{array} \right] \frac{0.433 \text{ PSI/Ft.}}{(4)} = \frac{0.440 \text{ PSI/Ft.}}{(4)} \left[\begin{array}{c} \text{Well Fluid} \\ \text{Composite} \\ \text{Gradient} \end{array} \right]$$

1)

c. Fluid Over Pump (FOP)

$$1) \left[\begin{array}{c} \text{Pump Intake} \\ \text{Pressure} \end{array} \right] \frac{500 \text{ PSI}}{1} \div \left[\begin{array}{c} \text{Well Fluid} \\ \text{Composite} \\ \text{Gradient} \end{array} \right] \frac{0.440 \text{ PSI/Ft.}}{(4)} = \frac{1136 \text{ Ft.}}{(5)} \left[\text{FOP} \right]$$

1)

d. Vertical Lift H_D

$$1) \left[\frac{\text{Pump Vertical}}{\text{Setting Depth-}V_D} \right] \frac{9000 \text{ Ft.}}{(5)} - \left[\frac{\text{Fluid Over}}{\text{Pump-FOP}} \right] \frac{1136 \text{ Ft.}}{(5)} = \frac{7864 \text{ Ft.}}{(6)} \left[\frac{\text{Vertical}}{\text{Lift-}H_D} \right] \quad 1)$$

e. Tubing Friction Loss - H_F (Ref. Fig. 2)

$$1) \left[\frac{\text{Gross}}{\text{Production}} \right] \frac{1100 \text{ B/D in}}{(7)} \left[\frac{\text{Nom.}}{\text{Tubing}} \right] \frac{2\%}{(7)} = \frac{33 \text{ Ft./1000}}{(7)} \left[\frac{\text{Friction}}{\text{Loss Per}} \right] \frac{1000 \text{ Ft.}}{(7)}$$

$$2) \left[\frac{\text{Friction}}{\text{Loss Per}} \right] \frac{33 \text{ Ft./1000}}{(7)} \times \left[\frac{\text{Tubing}}{\text{Length}} \right] \frac{9000}{(8)} = \frac{297 \text{ Ft.}}{(8)} \left[\frac{\text{Tubing}}{\text{Friction}} \right] \frac{\text{Loss-}H_F}{(8)}$$

1) - 2)

f. Producing Tubing Head - H_T (Ft.)

$$1) \left[\frac{\text{Producing}}{\text{Tubing}} \right] \frac{100 \text{ PSI}}{(9)} \div \left[\frac{\text{Composite}}{\text{Gradient}} \right] \frac{0.440}{(4)} = \frac{227 \text{ Ft.}}{(9)} \left[\frac{\text{Producing}}{\text{Tubing}} \right] \frac{\text{Head-}H_T}{(9)} \quad 1)$$

$$\left[\frac{\text{Vertical}}{\text{Lift-}H_D} \right] \frac{7864 \text{ Ft.}}{(6)} + \left[\frac{\text{Tubing}}{\text{Friction}} \right] \frac{297 \text{ Ft.}}{(8)} + \left[\frac{\text{Producing}}{\text{Tubing}} \right] \frac{227 \text{ Ft.}}{(9)} = \frac{8388 \text{ Ft.}}{(10)} \left[\frac{\text{Total}}{\text{Dynamic}} \right] \frac{\text{Head TDH}}{(10)}$$

h. Now Select Pump for 1100 B/D to Fit in 5½" Casing

- 1) A review of the performance (Head-capacity) curves results in a choice of pump type A (Exhibit 1.2), which shows maximum efficiency (66%) at required capacity (1100 B/D).

i. Determine Lift (Head) Per Stage

- 1) Referring to Exhibit 1.2 read vertically from 1100 B/D on the capacity scale to the head-capacity curve intersection.
- 2) From intersection read horizontally to left to head in feet per stage = 29.0 Ft./₍₁₁₎.

j. Determine Horsepower Required Per Stages

- 1) Referring to Exhibit 1.2 read vertically from 1100 B/D on the capacity scale to the Brake

Horsepower Curve intersection. NOTE: This curve is for a specific gravity of 1.0.

- 2) From intersection read horizontally to right to Brake Horsepower per stage of 0.35 H.P. While the horsepower per stage at the required capacity (0.35 HP/Stage) at 1100 B/D can be used, it is more common to use the maximum horsepower per stage factor. For the Type A pump stage = 0.38 HP/Stage₍₁₂₎

CAUTION: Some performance curves are for single stages and others for 100 stages.

k. Number of Pump Stages Required

$$1) \left[\frac{\text{Total Dynamic Head (TDH)}}{\text{Head (TDH)}} \right] \frac{8388 \text{ Ft.}}{(10)} \div \left[\frac{\text{Head Per Stage}}{\text{Stage}} \right]$$

$$\frac{29.0 \text{ Ft./Stage}}{(11)} = \frac{289 \text{ Stages}}{(13)} \quad 1)$$

- 2) Refer to Exhibit 1.3, Pump Specifications, to find closest number of stages in a standard pump housing. For the Type A pump ²⁸⁸/₍₁₄₎ stages would be selected. It is Housing No. 17T, is 29.2 feet long and weighs 863 pounds.

I. Motor Horsepower Required

- 1) Determine pump chamber horsepower.

$$\left[\frac{\text{No. of Pump Stages}}{\text{Stages}} \right] \frac{288}{(14)} \times \left[\frac{\text{HP Per Stage}}{\text{Stage}} \right] \frac{0.38 \text{ HP/Stage}}{(12)}$$

$$\times \left[\frac{\text{Water* Sp. Gr.}}{\text{Sp. Gr.}} \right] \frac{1.060}{(15)} = 116 \text{ H.P.}$$

(*For high water cut wells use produced water specific gravity. For low water cut wells use composite well fluid specific gravity.)

- 2) Referring to motor seal section specifications (Exhibit 1.5) determine horsepower required by motor seal chamber. It is 2.5 H.P.
- 3) Determine motor horsepower required by adding pump brake horsepower, motor seal section horsepower and gas separator (if used) horsepower.

$$[116 + 2.5 + 0 = \frac{118.5}{(15)}]$$

- 4) Referring to Motor Rating Specifications (Exhibit 1.4) the closest motor size is ¹²⁰/₍₁₆₎ H.P. It is

29.5 feet long and weighs 1430 pounds.

5) Check fluid flow velocity for motor cooling (Fig. 19). Since this is a 4½" O.D. motor in 5½" casing, the velocity at 1100 B/D is well above the 1.0 ft./sec. minimum recommended velocity.

6) A choice of two motors is available:

a) 1150 V. and 67 A.

b) $^{2280 \text{ V.}}_{(17)}$ and $^{34 \text{ A.}}_{(18)}$

In general, the higher voltage, lower amperage motor is the better choice in most installations, since it minimizes power cable losses. The choice of motor voltage may be based on a number of factors including well depth, casing size, horsepower size, cable size required, initial equipment cost and operating cost. Also, the power distribution voltage may be a factor.

m. Motor Seal Chamber

1) Referring to Exhibit 1.5 the Model 1 motor seal chamber is for use with single motors for 5½" casing. It is 4½ feet long and weighs 110 pounds.

n. Power Cable Length

$\left[\begin{array}{c} \text{Measured} \\ \text{Depth-MD} \end{array} \right] 9000 \text{ Ft.} + 100 \text{ Ft.} =$ 19100 is the measured pump setting depth (MD) plus 100 feet.

9100 Ft. $\left[\begin{array}{c} \text{Power} \\ \text{Cable} \\ \text{Length} \end{array} \right]$

o. Power Cable Size Selection

1) Using motor name plate voltage (2280V.) calculate recommended cable voltage drop.

$$\frac{2280 \text{ V.}}{(17)} \times \left[\begin{array}{c} \text{Voltage} \\ \text{Drop} \\ \text{Factor} \end{array} \right] \frac{0.05^*}{(18)} =$$

$$\frac{114 \text{ V.}}{(19)} \left[\begin{array}{c} \text{Cable} \\ \text{Voltage} \\ \text{Drop} \end{array} \right]$$

2) Now determine voltage drop per 1000 ft.

$$\left[\begin{array}{l} \text{Cable} \\ \text{Voltage} \\ \text{Drop} \end{array} \right] \frac{114\text{V.}}{(19)} \div \left[\begin{array}{l} \text{Cable} \\ \text{Length} \end{array} \right] \frac{9100 \text{ Ft.}}{1000} =$$

$$\frac{12.5\text{V.}/1000 \text{ Ft.}}{(20)} \left[\begin{array}{l} \text{Recommended} \\ \text{Cable} \\ \text{Voltage} \\ \text{Drop}/ \\ 1000 \text{ Ft.} \end{array} \right]$$

3) Reference Fig. 12 (Voltage Drop Chart) for operating Bottom Hole temperature (190°F), find the conductor Voltage Drop Correction factor = $1.267_{(12)}$

4) Now divide the

$$\left[\begin{array}{l} \text{Recommended} \\ \text{Cable Voltage} \\ \text{Drop}/1000 \text{ Ft.} \end{array} \right] \frac{12.5\text{V.}}{(20)} \div \left[\begin{array}{l} \text{Voltage} \\ \text{Drop} \\ \text{Correction} \\ \text{Factor} \end{array} \right]$$

$$\frac{1.267}{(21)} = \frac{9.9/1000 \text{ Ft.}}{(22)}$$

Now enter Fig. 12 at 9.9V./1000 Ft. on left hand scale and read horizontally to the right to motor amperage (34 amps) and choose cable size at or below this point. For 9.9V./1000 Ft. and 34 amps this would be a #2 Copper Cable₍₂₃₎ which results in an actual voltage drop of 9V./1000 Ft.

5) Check for motor start-up.

a) Determine actual operating motor voltage drop

$$\frac{9.0\text{V.}/1000 \text{ Ft.} \times 9100 \text{ Ft.} \times 1.267}{\left[\begin{array}{l} \text{Cable} \\ \text{Temp} \\ \text{Factor} \end{array} \right]} = 104\text{V.} \left[\begin{array}{l} \text{Operating} \\ \text{Voltage} \\ \text{Drop} \end{array} \right]$$

b. Determine in-rush voltage drop (4.0 factor considers sag in primary system and other system losses)

$$104\text{V.} \times \frac{4.0}{\left[\begin{array}{l} \text{In Rush} \\ \text{Voltage} \\ \text{Drop Factor} \end{array} \right]}$$

$$= 416\text{V.} \left[\begin{array}{l} \text{In Rush} \\ \text{Voltage} \\ \text{Drop} \end{array} \right]$$

c) Determine motor terminal start-up voltage.

$$2280\text{V.} - 416\text{V.} = 1864\text{V.} \left[\begin{array}{c} \text{Start-up} \\ \text{Motor} \\ \text{Terminal} \\ \text{Voltage} \end{array} \right]$$

d) Calculate start-up voltage at motor as percent of motor name plate voltage.

$$\frac{1864\text{V.}}{2280\text{V.}} \times 100 = 81.8\%$$

Since this is greater than 50% of motor name plate voltage motor start-up should not be a problem. The 50% minimum requirement and the 4.0 In Rush voltage drop factor are determined by operating experience. Some tests have indicated higher in-rush currents than 4.0.

Manufacturer should be contacted for specific recommendations.

**NOTE: The 5% voltage drop is based on NEC Article 215.2, and results in a reduced power loss and extended power cable life.*

Other limiting methods such as 30V/1000 ft. may be used to determine voltage loss in the power cable. The amount of voltage drop allowed must consider the ability to start the motor.

p. Power Cable Clearance

- 1) Having selected the power cable size required (#2 Copper Cable)₍₂₃₎, it is necessary to be certain that there is sufficient clearance for the power cable and equipment in the well bore.
- 2) The following power cable clearance calculation assumes that an offset pump discharge tubing head is used on the pump. Therefore, the well diameter (casing I.D.) must be greater than the largest of the following:

a) $A + B$

b) $\left(\frac{E - D}{2}\right) + \frac{A + B}{2}$

c) $\left(\frac{C - D}{2}\right) + \frac{A + B}{2}$

d) $\frac{C + E}{2} + F + G$

e) $\frac{H + E}{2} + F + G$

Where:

A = Tubing Coupling Diameter

B = Power Cable Diameter

C = Pump Diameter*

D = Unit Centerline/Discharge Head
Centerline Offset Dimension

E = Motor Diameter*

F = Motor Lead Extension Thickness

G = Band Thickness

H = Motor Seal Chamber

*Must include special coating thickness if used.

Therefore for this problem:

A = 3.063" (2% EU Coupling O.D.)

B = 1.320" (from Exhibit 1.10)

C = 4.000" (from Exhibit 1.10)

D = 0.220" (from Exhibit 1.12)

E = 4.500" (from Exhibit 1.10)

F = 0.450" (from Exhibit 1.11)

G = 0.030

H = 4.00" (from Exhibit 1.10)

Then making calculations:

a) $A + B = 3.063" + 1.320" = \frac{4.383"}{(24)}$

b) $\left(\frac{E - D}{2}\right) + \frac{A + B}{2} = \frac{4.500 - 0.220}{2} + \frac{3.063" + 1.320"}{2} = \frac{4.882"}{(25)}$

$$\text{c) } \left(\frac{C - D}{2} \right) + \frac{A + B}{2} = \frac{4.000 - 0.220}{2} + \frac{3.063}{2} + 1.320 = \frac{4.632}{2} \quad (26)$$

$$\text{d) } \frac{C}{2} + \frac{E}{2} + F + G = \frac{4.000}{2} + \frac{4.500}{2} + 0.450 + 0.030 = 4.730$$

$$\text{e) } \frac{H}{2} + \frac{E}{2} + F + G = \frac{4.00}{2} + \frac{4.50}{2} + 0.450 + 0.03 = 4.73$$

Therefore b) is largest = 4.882".

Then comparing it with the casing I.D. for clearance.

Casing I.D. (5½ × 17#) = 4.892"

$$\text{Equipment "Diameter" b) } = \frac{-4.882}{0.010} \quad (27)$$

Therefore the equipment and power cable "bundle" will go in the casing. However, extreme care should be taken due to this small clearance.

NOTE: *Exhibit 1.10* has a table for maximum cable/equipment size combinations.

q. Power Cable Insulation

1) Select power cable insulation based on well bore temperature at pump setting depth, motor current, and well bore environment. Using *Fig. 14* (for #2 power cable) find the well bore temperature (190°F) on left hand scale, and read horizontally to motor current (34 amps) intersection. This gives a conductor temperature of 200°F.

2) Since conductor temperature is 200°F select the proper power cable insulation from cable supplier's information.

CAUTION: *Some cable insulations can deteriorate when coming into contact with some types of corrosion inhibitors.*

r. Motor Lead Extension

1) Cable size of the motor lead extension and motor potlead size is determined by the annular space available. Choose the largest size available that will provide acceptable clearance in the well bore.

2) Next, select the type of armor that is appropriate for the well bore conditions. Available materials may include galvanized steel, phosphur bronze, and monel. Phosphur bronze armor will be chosen since this well has no corrosion problem.

3) The motor lead extension length is determined by making it reach at least 5 feet beyond the pump discharge head. This is calculated as follows:

- (a) Pump Length 29.2 Ft.
- (b) Motor Seal Chamber..... 4.5 Ft.
- (c) Length Above Pump..... +5.0 Ft.
- (d) Additional Accessory Equipment ... 5.0 Ft.
- (e) Motor Lead Extension Length $43.7 \text{ Ft.} / (29)$

Referring to **Exhibit 1.11** a 60 foot motor lead extension is available and is chosen for this installation.

(4) A motor lead extension cable guard should be chosen to go with this.

s. Cable Bands

- 1) Stainless steel bands were requested by user so these will be provided.
- 2) One 25 count bundle of 30" bands is required for securing the motor lead extension.
- 3) Two 22" bands per tubing joints are required or:

$$\frac{9000 \text{ Ft.}}{30 \text{ Ft./Joint}} \times 2 \text{ Bands/Joint} = \frac{600 \text{ Bands}}{(30)}$$

t. Wellhead

1) Select a wellhead sufficient to carry the weight of the tubing, equipment and fluid.

- (a) Motor..... = 1430 lbs.
- (b) Motor Seal Chamber..... = 110 lbs.
- (c) Pump..... = 863 lbs.
- (d) Tubing 4.7 lb/ft. × 9000 ft. = 42,300 lbs.
- (e) Power Cable 1.62 lb/ft. × 9000 ft. = 14,580 lbs.
- (f) Motor Lead 1.70 lb/ft. × 40 ft. = 70

(g) Cable Bands $15 \text{ lb}/100 \times 600 \dots\dots\dots = 90$

(h) Fluid = $1.060 \text{ sp. gr.} \times 8.34 \text{ lb}/\text{gal} \times 0.1626 \text{ gal}/\text{ft.} \times 9000 \text{ ft.} = \underline{12,940}$

(i) Total Weight on Tubing Head $\dots\dots\dots = 72,383 \text{ lbs.}/_{(31)}$

2) Based on this total weight and referring to Exhibit 1.8, the Type B tubing head for 5½" O.D. casing size by 23/8" tubing size with a pack-off for #2 round cable is chosen.

3) $288 \text{ stages} \times 29 \text{ ft.}/\text{stage} \times 0.440 = 3675 \text{ psi}$. Therefore all valves and fittings on the tubing must be rated greater than 3675 psi.

u. Motor Control Panel and Junction Box

1) Calculate surface "no load" voltage:

(a) Motor Name Plate Voltage = $2280\text{V}/_{(17)}$.

(b) Cable Voltage Drop = $+104\text{V}/_{(19)}$.

(c) Surface Full Load Voltage = $2284\text{V}/_{(32)}$

(d) Transformer Drop Factor (3%) X1.03

(e) Surface "No Load" Voltage $2456\text{V}/_{(33)}$.

2) Referring to Exhibit 1.6 a motor control panel is selected which has adequate voltage and amp rating to accommodate system voltage and current (2456 V. and 34 amps). Therefore, the Size D3 MCP is selected. Various control options are available. Lightning arrestors should be specified if applicable.

3) A junction box should be specified for the installation.

$$\left[\begin{array}{l} \text{Surface} \\ \text{Full} \\ \text{Load} \\ \text{Voltage} \end{array} \right] \frac{2384 \text{ V.}}{(32)} \times \left[\begin{array}{l} \text{Motor} \\ \text{Amps} \end{array} \right] \frac{34 \text{ A.}}{(18)} \times$$

in KVA is:

$$\frac{1.732}{1000} = \frac{140 \text{ KVA}}{(32)}$$

2) For 3 single phase transformers:

$$\frac{140 \text{ KVA}}{(34)} \div \left[\frac{\text{No. of}}{\text{Transformers}} \right] \frac{3}{3} = \frac{47 \text{ KVA}}{(35)}$$

Therefore, use three 50 KVA transformers. Referring to [Exhibit 1.7](#) for single phase transformers the 50 KVA for a primary voltage of 14,400/24,940 should be selected. As the note indicates, additional secondary voltages can be obtained by WYE connecting the secondary. This is how the 2400 V. system voltage is obtained.

5.2 Example Problem No. 2

— Water and Oil Production with Gas

5.2.1

The following exhibit sheets provide data and information required for Example Problem No. 2.

- a. [Exhibit 2.1](#)— Well Data Sheet.

5.2.2

Unless otherwise specified, data used in sizing calculations comes from the Well Data Sheet ([Exhibit 2.1](#)).

- a. From [Figure 17](#) determine gas solution (R_g) at 500 PSI PIP = 83 ft.³/bbl.
- b. Enter [Figure 18](#) at 83 ft.³/bbl. (R_g @ 500 PSI PIP) and determine formation volume factor (B_o) = 1.1 bbl/bbl. STO.
- c. Determine formation gas volume factor B_g at 500 PSI PIP as follows:

$$B_g = \left[\frac{.0283 \frac{ZT}{P}}{P} \right] \text{ ft. }^3/\text{scf} \text{ or } \left[\frac{.0283 \frac{ZT}{P}}{P} \right] \times$$

$$\left[\frac{1728 \times 1000}{231 \times 42} \right] = 5.04 \frac{ZT}{P} \text{ bbl/scf}$$

$$B_g = \frac{5.04 \times 0.85 \times 650}{515} = 5.4 \text{ bbl/MSCF}$$

Use gas compressibility factor (Z) = .85 (Note Z varies with gas Sp. gr., pressure and temperature. Z = .85 is used as an average).

- d. Determine if gas separator is required, that is, if free gas at PIP exceeds 10% by volume:

(1) Total gas in solution at bubble point = B/D oil $\times R_s$ =
 $\frac{220 \times 440}{1000} = 96.8 \text{ Mcf day}^*$

(2) Total gas still in solution at PIP = R_s
 $\times \text{B/D oil} = \frac{220 \times 83}{1000} = 18.26$
 Mcf day

(3) Gas liberated or Free gas = $96.8 - 18.26 = 78.54 \text{ Mcf day}^*$

*NOTE: If GOR is used instead of R_s this number is slightly larger.

(4) Volume of oil = $V_o = \text{BOPD @ PIP} \times B_o$
 $B_o = 220 \times 1.1 = 242 \text{ B/D}$
 Volume of gas = Free gas $\times B_g$ =
 $78.54 \times 5.4 = 424 \text{ B/D}$
 Volume of water =
 $V_w = \text{given} = 880 \text{ B/D}$
 $V_t = 1546 \text{ B/D}$
 $\frac{V_g}{V_t} = \frac{424 \times 100}{1546} = 27\% \text{ (27\% > 10\% therefore a suitable gas separator is required.)}$

e. Assume gas separator is 90% effective and 10% of gas is not separated or $V_g = 10\%$. Therefore, .1 $\times 424 = 42$ entering the pump.

f. Therefore the total fluid mixture in B/D entering the pump is:

$V_o =$	242 B/D
$V_g =$	42 B/D
$V_w =$	880 B/D
$V_t =$	1164 B/D

g. The amount of free gas entering the pump as a percent of total fluid is:

$$\frac{V_g}{V_t} = \frac{42 \times 100}{1164} = 3.6\%$$

While gas represents 3.6% by volume of the fluid being pumped it has no significant effect on the well fluid composite specific gravity.

h. Now that the total volume of the produced mixture through the pump is known the method of equipment selection is identical to that shown in Example Problem No. 1, Section 5.1.2.

5.3 Example Problem No. 3

5.3.1

The following exhibit sheets provide data and information required for Example Problem No. 3.

- a. **Exhibit 3.1**— Well Data Sheet.

5.3.2

Unless otherwise specified, data used in sizing calculations comes from the Well Data Sheet (**Exhibit 3.1**).

- a. Using **Fig. 3**, find the gas free crude viscosity for 15 deg. API crude at 140°F to be 150 Cp.
- b. Using **Fig. 4**, find the gas in solution at 300 PSI (PIP) to be 50 cu. ft./bbl.
- c. Using **Fig. 5**, find the gas-saturated crude oil viscosity for 150 Cp at a 50 cu. ft./bbl. gas in solution ratio to be approximately 70 Cp.

d. Well Fluid Composite Specific Gravity

1. Water cut 0.20 × Water Sp. Gr. 1.060

$$= \frac{.212}{(1)} \left[\begin{array}{c} \text{Water Phase} \\ \text{Sp. Gr.} \end{array} \right]$$

2. Oil Cut 0.80 × Oil Sp. Gr. .966

$$= \frac{.733}{(2)} \left[\begin{array}{c} \text{Oil Phase} \\ \text{Sp. Gr.} \end{array} \right]$$

3. Water Phase Sp. Gr. .212 + Oil Phase
(1)

$$\text{Sp. Gr. } \frac{.773}{(2)} = .985 \left[\begin{array}{c} \text{Well Fluid} \\ \text{Composite} \\ \text{Sp. Gr.} \end{array} \right]$$

1. - 3.

NOTE: Well fluid composite contains no free gas by volume.

- e. Using **Fig. 6**, convert 70 Cp to SSU using an average specific gravity of 0.985. Read 350 SSU.
- f. Using **Fig. 7**, apply the correction factor for a 20.0% water cut. Read 2.0.
- g. The 20.0% water cut increases the actual viscosity by a factor of two. Corrected viscosity would be 700 SSU.
- h. Refer to **Fig. 11** for correction factors. Since installation is in 5.50 O.D. casing, assume 60% efficient pump for required volume.

$$1. \text{ Capacity Factor (Qvis)} = 0.805$$

$$2. \text{ Head Factor (Hvis)} = 0.825$$

$$3. \text{ Horsepower Factor (bhp}_{vis}) = 1.30$$

The effects of viscosity on the unit have now been determined and the sizing of the unit can now proceed and the correction factors used where required.

i. Well Fluid Composite Gradient

$$1. \left[\begin{array}{c} \text{Well Fluid} \\ \text{Composite} \\ \text{Sp. Gr.} \end{array} \right] \frac{.985}{(3)} \times \left[\begin{array}{c} \text{Fresh Water} \\ \text{Pressure} \\ \text{Gradient} \end{array} \right]$$

$$\frac{0.433}{(4)} = \frac{0.427}{(4)} \left[\begin{array}{c} \text{Well Fluid} \\ \text{Composite} \\ \text{Gradient} \end{array} \right] \quad 1.$$

j. Fluid Over Pump (FOP)

$$1. \left[\begin{array}{c} \text{Pump} \\ \text{Intake} \\ \text{Pressure} \end{array} \right] \frac{300 \text{ PSI}}{(4)} \div \left[\begin{array}{c} \text{Corrected} \\ \text{Well Fluid} \\ \text{Composite} \\ \text{Gradient} \end{array} \right]$$

$$\frac{0.427}{(4)} = \frac{703 \text{ Ft.}}{(5)} \quad (\text{FOP}) \quad 1.$$

k. Vertical Lift — H_D

$$1. \left[\begin{array}{c} \text{Pump Vertical} \\ \text{Setting Depth} \end{array} \right] (V_D) \quad \underline{5200 \text{ Ft.}}$$

$$- \text{Fluid Over Pump} \quad \frac{703 \text{ Ft.}}{(5)} = \frac{4497}{(6) \text{ Ft.}} \left[\begin{array}{c} \text{Vertical} \\ \text{Lift H}_D \end{array} \right] \quad 1.$$

l. Vertical Lift Corrected for Viscosity — H_D-vis

$$1. \left[\begin{array}{c} \text{Net} \\ \text{Lift} \end{array} \right] \frac{4497}{(6)} \div \left[\begin{array}{c} \text{Viscosity} \end{array} \right] \frac{0.825}{(6)} =$$

$$\frac{5451 \text{ Ft.}}{(7)} \left[\begin{array}{c} \text{Corrected} \\ \text{Vertical} \\ \text{Lift} \end{array} \right] \quad 1.$$

m. Tubing Friction Loss — H_F (Ref. Fig. 8)

For Proper Selection for the Viscosity of the Fluid to be Pumped.

$$1. \left[\begin{array}{c} \text{Gross} \\ \text{Production} \\ \text{Rate} \end{array} \right] \frac{800 \text{ B/D in.}}{\left[\begin{array}{c} \text{NOM.} \\ \text{Tubing} \\ \text{Size} \end{array} \right]}$$

$$\frac{2\frac{3}{8}''}{(8)} = \frac{145 \text{ Ft./1000}}{(8)} \left[\begin{array}{c} \text{Friction} \\ \text{Loss Per} \\ \text{1000 Ft.} \\ \text{At 700 SSU} \end{array} \right]$$

$$2. \left[\begin{array}{c} \text{Friction} \\ \text{Loss Per} \\ \text{1000 Ft.} \end{array} \right] \frac{145 \text{ Ft./1000}}{(7)} \times \left[\begin{array}{c} \text{Tubing} \\ \text{Length} \end{array} \right]$$

$$1. \left[\begin{array}{c} \text{Producing} \\ \text{Tubing} \\ \text{Pressure} \end{array} \right] \frac{50 \text{ PSI}}{\left[\begin{array}{c} \text{Composite} \\ \text{Gradient} \end{array} \right]} \frac{\left[\begin{array}{c} \text{Tubing} \\ \text{Length} \end{array} \right]}{1000 \text{ Ft.}} \quad 1. - 2.$$

$$\frac{0.427}{(4)} = \frac{117 \text{ Ft.}}{(10)} \left[\begin{array}{c} \text{Producing} \\ \text{Tubing Head} \\ H_T \end{array} \right] \quad \text{— } H_T \text{ (ft.)}$$

1. - 2.

$$2. \left[\begin{array}{c} \text{Producing} \\ \text{Tubing} \\ \text{Head } H_T \end{array} \right] \frac{117 \text{ Ft.}}{(10)} \div \left[\begin{array}{c} \text{Viscosity} \\ \text{Correction} \\ \text{Factor} \end{array} \right]$$

$$\frac{0.825}{(11)} = \frac{142 \text{ Ft.}}{(11)} \left[\begin{array}{c} \text{Corrected Tubing} \\ \text{Head } H_T \end{array} \right] \quad 1. - 2.$$

o. Pump Total Dynamic Head (Ft.)

$$\left[\begin{array}{c} \text{Corrected} \\ \text{Vertical} \\ \text{Lift-} H_D \end{array} \right] \frac{5451 \text{ Ft.}}{(7)} + \left[\begin{array}{c} \text{Friction Loss} \\ \text{Per 1000 Ft.} \\ \text{At 700 SSU} \end{array} \right] \frac{754 \text{ Ft.}}{(9)} +$$

$$\left[\begin{array}{c} \text{Corrected} \\ \text{Tubing} \\ \text{Head-} H_T \end{array} \right] \frac{142 \text{ Ft.}}{(11)} = \frac{6347 \text{ Ft.}}{(11)} \left[\begin{array}{c} \text{Total} \\ \text{Dynamic} \\ \text{Head-TDH} \end{array} \right]$$

p. Correct 800 B/D Required Rate for Viscosity @ 700 SSU

$$1. \left[\frac{\text{Required}}{\text{STB}} \right] \frac{800 \text{ B/D}}{\div} \left[\frac{\text{Capacity}}{\text{Correction}} \right] \left[\frac{\text{Factor}}{\text{Factor}} \right]$$

$$\frac{.805}{(13)} = \frac{994 \text{ B/D}}{\text{Volume}} \left[\frac{\text{Corrected}}{\text{Volume}} \right]$$

1.

q. Select Pump for 994 B/D to Fit in 5½" Casing

1. A review of the performance (head-capacity) curves results in a choice of pump Type B (Exhibit 1.2), which shows efficiency (65%) at required capacity (994 B/D).

r. Determine Lift (Head) Per Stage

1. Referring to Exhibit 1.2 read vertically from 994 B/D on the capacity scale to the head-capacity curve intersection.
2. From intersection read horizontally to left to head in feet per stage = 30.0 Ft./(14)

s. Determine Horsepower Required Per Stage

1. Referring to Exhibit 1.2 read vertically from 994 B/D on the capacity scale to the brake horsepower curve intersection.
2. From intersection read horizontally to right to brake horsepower per stage of 0.28 H.P. maximum horsepower/stage is stage frequently used.

t. Number of Pump Stages Required

$$1. \left[\frac{\text{Total}}{\text{Dynamic}} \right] \left[\frac{\text{Head-TDH}}{\text{Head-TDH}} \right] \frac{6347}{(12)} \div \left[\frac{\text{Head Per}}{\text{Stage}} \right]$$

$$\frac{23.5 \text{ Ft./Stage}}{(14)} = \frac{212}{(15)}$$

1.

u. Pump Brake Horsepower Required

$$1. \left[\frac{\text{No. of}}{\text{Pump}} \right] \left[\frac{\text{Stages}}{\text{Stages}} \right] \frac{270 \times \text{HP}}{(15)} \frac{0.28 \text{ HP/Stage} \times}{(14)}$$

$$\left[\frac{\text{Horsepower}}{\text{Viscosity}} \right] \frac{1.30 \times \left[\frac{\text{Sp.}}{\text{Gr.}} \right] 0.985}{(16)} = \frac{103 \text{ HP}}{(16)}$$

1.

SECTION 6 SPECIAL WELL CONDITIONS

6.1 High Bottom Hole Temperature

— Generally, electric submersible pump installations in wells with bottom hole temperatures up to 200°F are handled as standard installations without special considerations.

6.1.1

For wells with bottom hole temperatures between 200-250°F special consideration must be given to:

a. Power Cable — general cable insulation material temperature limits can be categorized as follows:

Insulation Material	Well Bore Temp. °F	Conductor Temp. °F
1. Polyethylene (PE) Unarmored	140	167
2. PPE with Nitrile Jacket	170	200
3. EPR with Nitrile Jacket	257	300

These are general guidelines that can be used when evaluating temperature limits of power cable.

The cable manufacturer should be consulted to determine specific recommendations for given well conditions.

b. Electric Submersible Pump Unit — Most manufacturers' standard equipment is capable of operation to 250°F well bore bottom hole temperature.

6.1.2

For wells with wellbore bottom hole temperature above 250°F special consideration must be given to:

a. Power Cable — Lead sheathed flat power cable should be considered for these applications. There may be other cables also available and the power cable manufacturers should be consulted.

b. Electric Submersible Pump Unit — For operation above 250°F it may be necessary for certain modifications to be made to the electric submersible pump unit, such as elastomers, insulating materials and motor oil, or other design and material changes.

6.2 Corrosion

— If general "oilfield corrosion" is encountered normal corrosion resistant materials and coatings are available to provide protection.

6.2.1 Power Cable and Banding Material

- a. Galvanized or monel armored cable is available for protection in most corrosive environments.
- b. In high H₂S corrosion environments it may be necessary to go to a lead sheathed cable to prevent copper deterioration.
- c. Banding materials in stainless steel or monel are available for corrosive environments.

6.2.2 Electric Submersible Pump Unit

- a. Coatings for corrosion resistance are available and range from a standard paint, special plastic coating, or spray metal coatings such as monel. If special coatings are specified the coating thickness must be included in the clearance calculations (i.e., Section 5.1.2 p.).
- b. Some corrosive conditions may require special metallurgy for the housing or other components. The pump manufacturer should be contacted to determine what changes may be required.
- c. Anodes are also available for connection to the bottom of the motor.

6.3 Solids

6.3.1

If solids are expected to be a problem in an installation the well bore should be cleaned out prior to installing equipment.

6.3.2

Special considerations can be made by the pump manufacturer to improve the ability of the pump to handle solids. These may include both material and design changes. The pump manufacturer should be consulted relative to specific applications.

SECTION 7 OTHER ELECTRIC SUBMERSIBLE PUMP EQUIPMENT CONSIDERATIONS

7.1 Electric Submersible Pump variable speed control and soft start.

7.1.1 Variable Speed Control

— Allows the Electric Submersible Pump speed to be varied above or below normal operating speed in order to produce the well most effectively. The speed change may be necessary to determine productivity on a new well or to meet changing conditions in an existing well.

a. The variable speed drive takes the place of the motor controller and incorporates all of the normal regulations and control usually found in a motor controller.

b. By varying pump speed of the centrifugal downhole pump the affinity laws apply as follows:

$$\begin{aligned} 1) \text{ Flow: } Q_2 &= Q_1 \left(\frac{N_2}{N_1} \right) \\ 2) \text{ Head: } H_2 &= H_1 \left(\frac{N_2}{N_1} \right)^2 \\ 3) \text{ Brake H.P.: } BHP_2 &= BHP_1 \left(\frac{N_2}{N_1} \right)^3 \end{aligned}$$

Where:

Q_1 , H_1 , BHP_1 , N_1 = Initial capacity,
Head, Brake Horsepower and speed,
respectively.

1) - 3)

Q_2 , H_2 , BHP_2 , N_2 = New capacity,
Head, Brake Horsepower and speed,
respectively.

1) - 3)

From the above relationships it can be seen that brake horsepower limitations may be quickly reached with increased speed or that insufficient head (shut off) may be reached with reduced speed.

c. The sizing and application of the variable speed drive is outside the scope of this RP. Equipment manufacturer should be consulted.

7.1.2 Soft Start

— Equipment that provides a means of limiting in-rush current and voltage intended to reduce electrical and mechanical stresses at start-up in order to improve unit life.

a. Limiting in-rush current reduces power line voltage losses which may be detrimental to other equipment on line.

b. Provides less stress on the power cable during start-up

c. Reduces high starting torques that occur on across the line starting, therefore putting lower electrical and mechanical stress on the downhole pump and motor.

d. The sizing and application of the soft start unit is outside the scope of this RP. Contact the equipment manufacturer for this information.

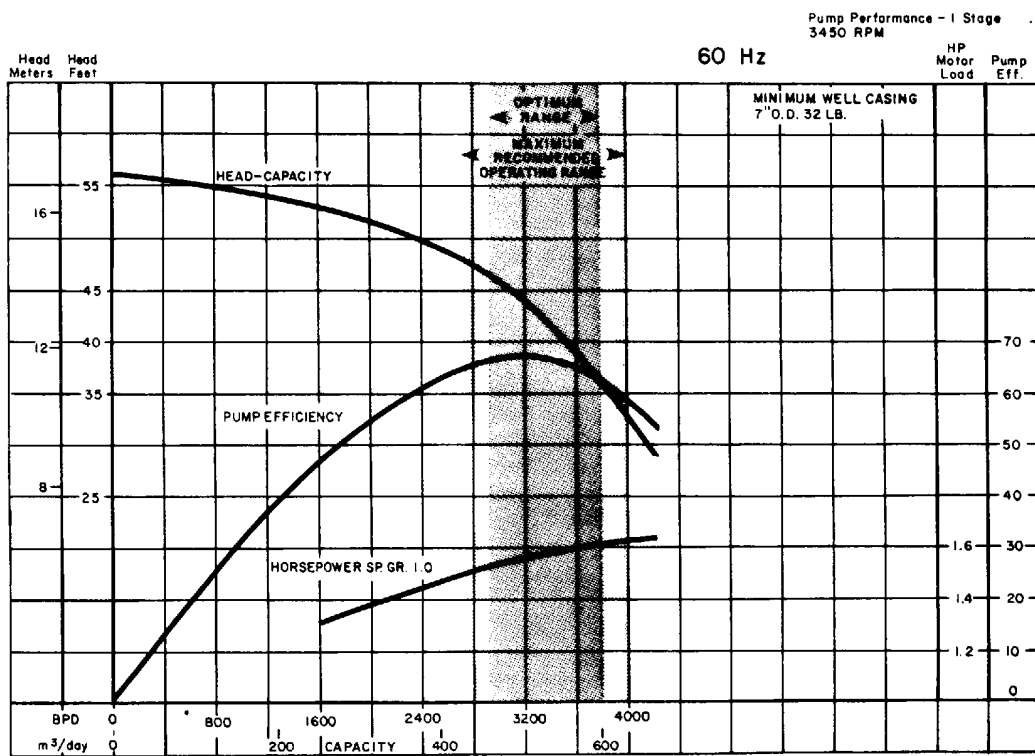
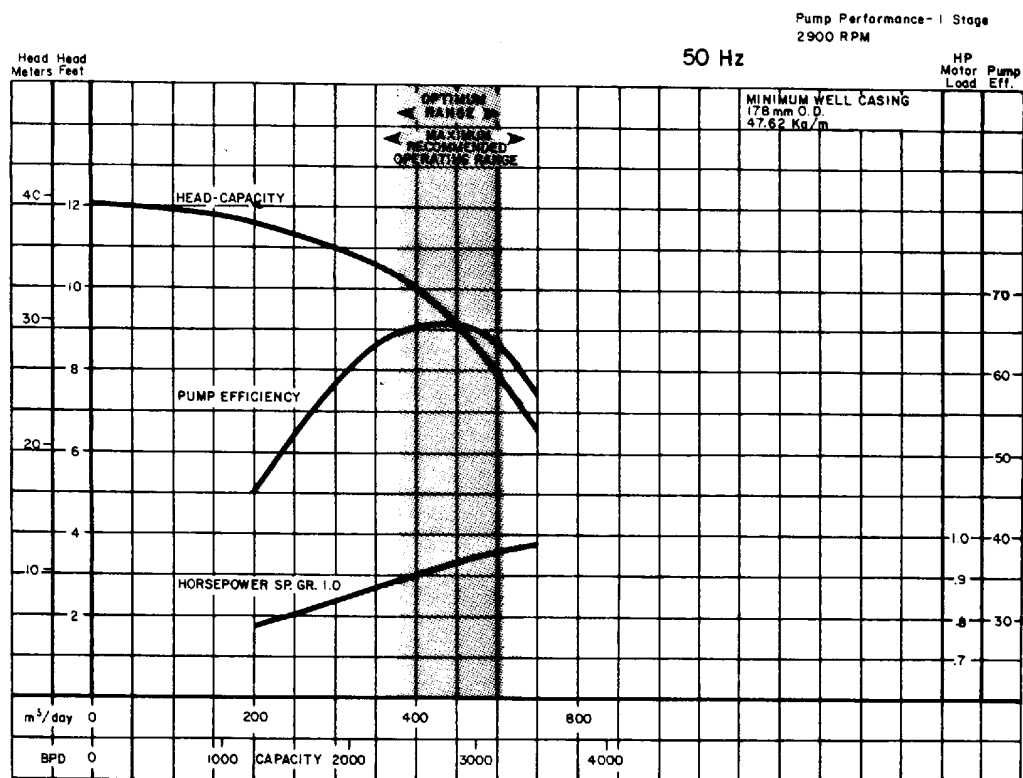


FIG. 1
PUMP PERFORMANCE CURVE

FIG. 1 PUMP PERFORMANCE CURVE

FLOW LOSSES DUE TO FRICTION IN A.P.I. PIPE

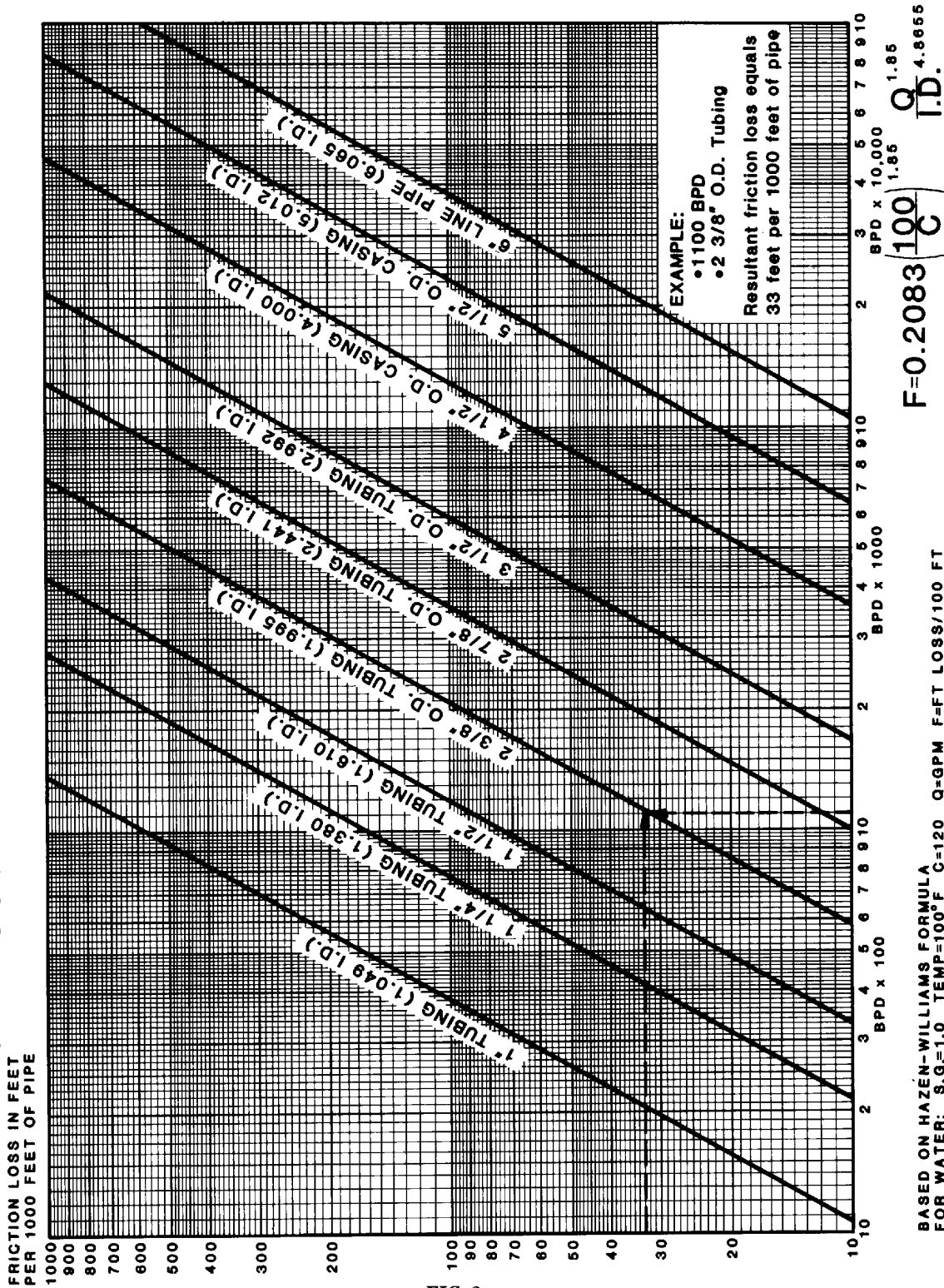


FIG. 2
FLOW LOSSES DUE TO FRICTION IN API PIPE

FIG. 2 FLOW LOSSES DUE TO FRICTION IN API PIPE

VISCOSITY OF GAS-FREE CRUDE OIL AT OIL FIELD TEMPERATURES

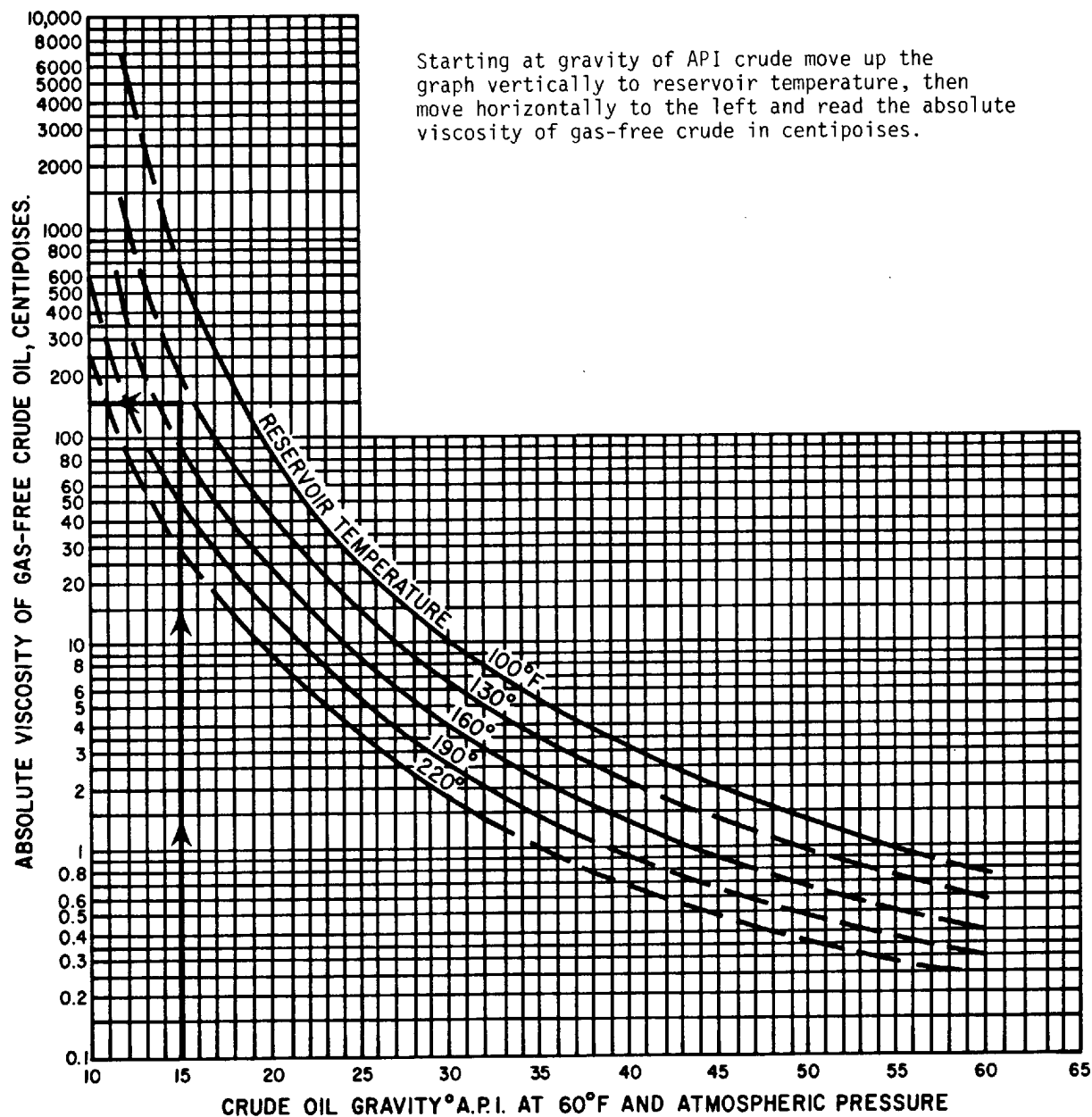


FIG. 3
VISCOSITY OF GAS-FREE CRUDE OIL AT OIL FIELD TEMPERATURES

FIG. 3 VISCOSITY OF GAS-FREE CRUDE OIL AT OIL FIELD TEMPERATURES

From operating pressure move up vertically to API oil gravity, move horizontally to left and read gas in solution value.

PREDICTION OF SOLUBILITY FROM SATURATION PRESSURE AND CRUDE OIL GRAVITY

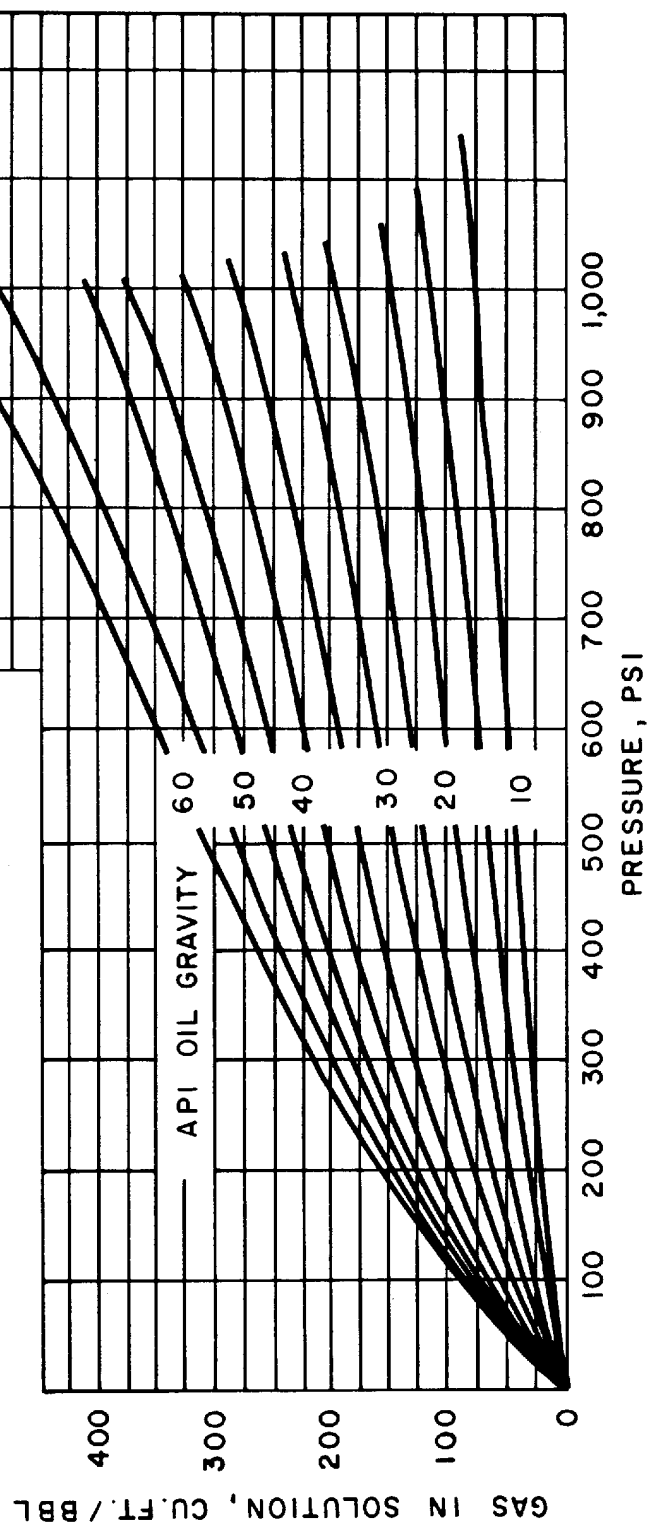


FIG. 4

FIG. 4 PREDICTION OF SOLUBILITY FROM SATURATION PRESSURE AND CRUDE OIL GRAVITY

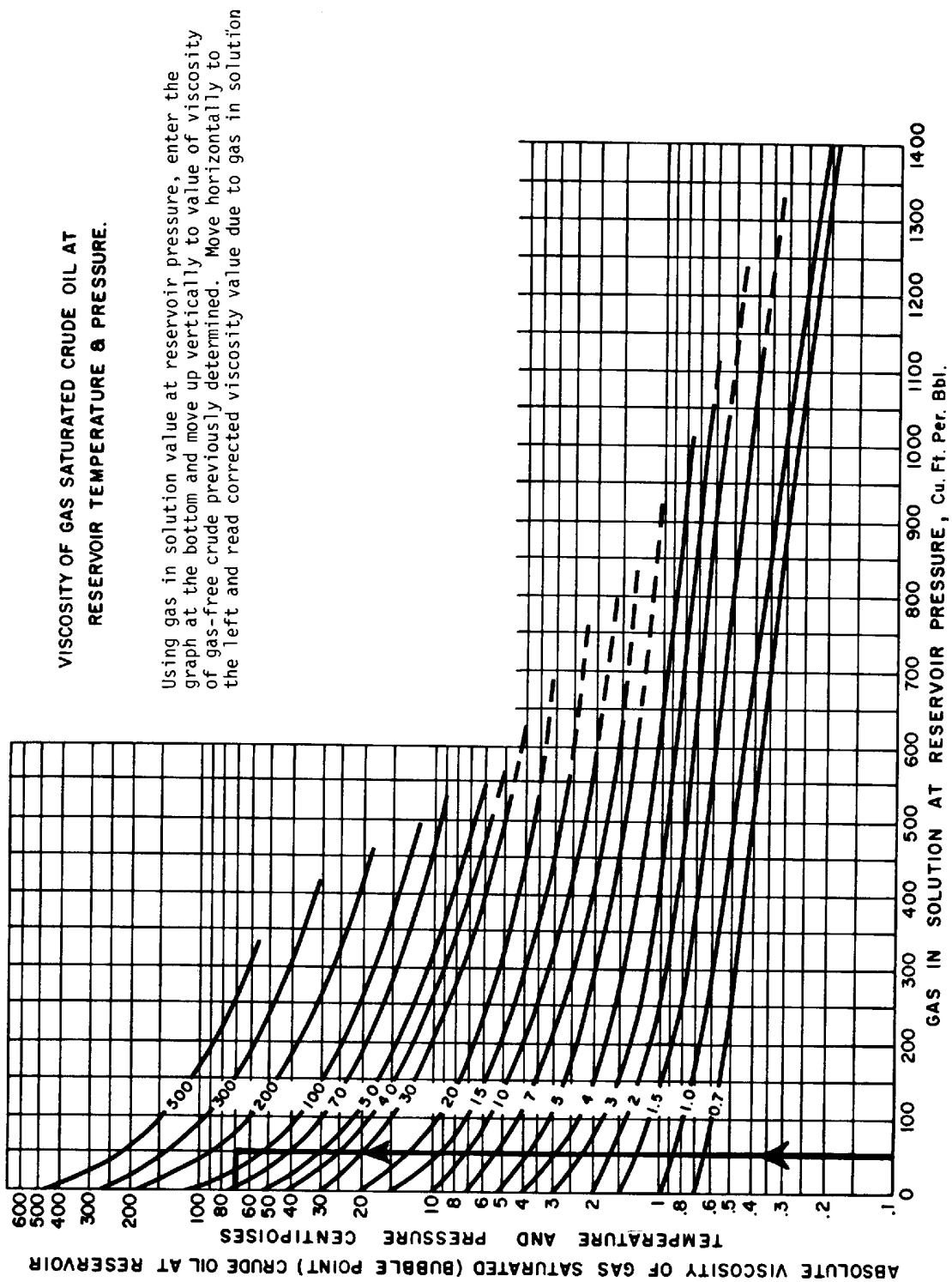
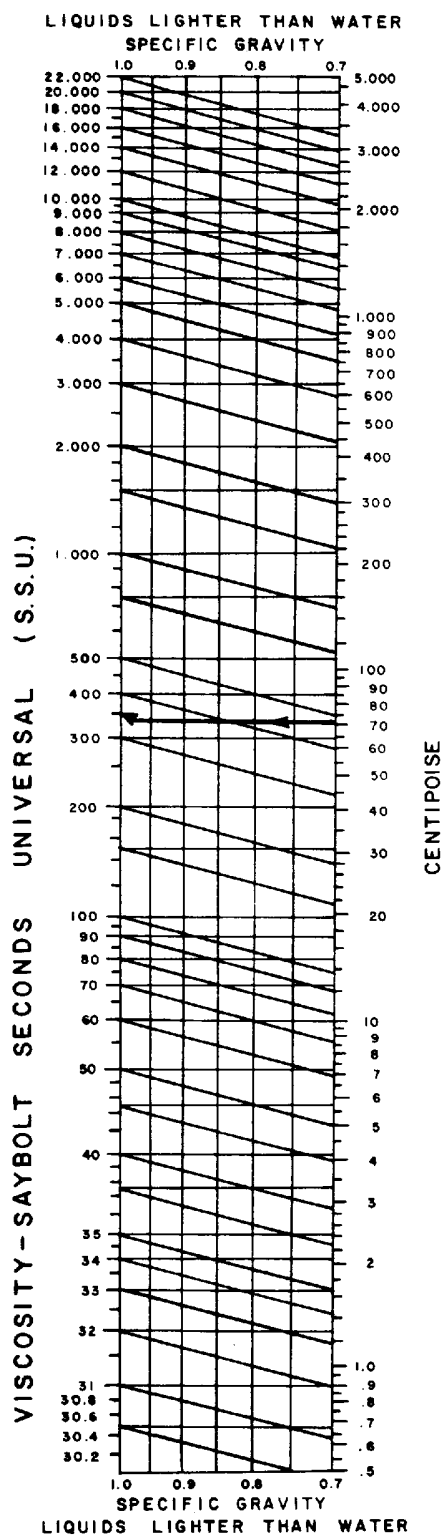


FIG. 5 VISCOSITY OF GAS SATURATED CRUDE OIL AT RESERVOIR TEMPERATURE & PRESSURE

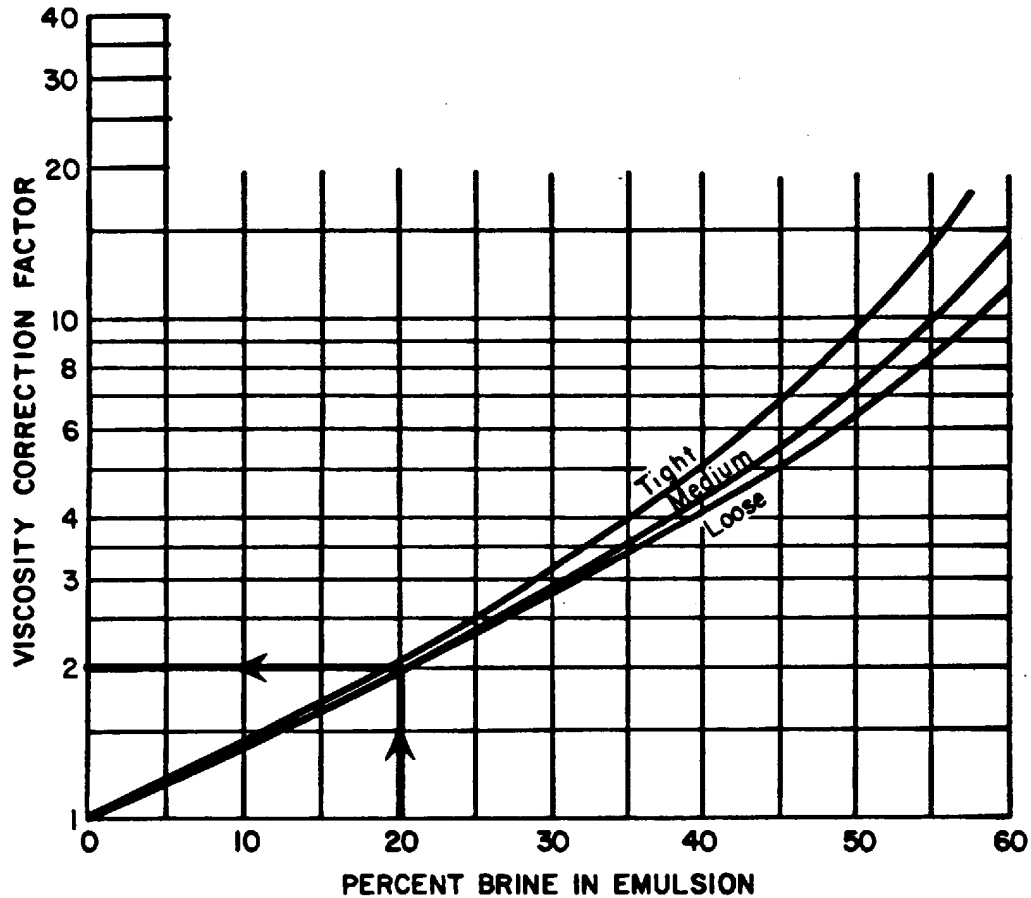


Using gas in solution viscosity value in centipoise, enter the graph at right and move horizontally to composite specific gravity value. From this point, move parallel to the diagonal line and read viscosity in SSU on the left scale.

FIG. 6

FIG. 6 VISCOSITY-SAYBOLT SECONDS UNIVERSAL (S.S.U.)

Using percent water cut, go up vertically on graph to type of emulsion, then horizontally to left. Read viscosity correction factor.



Caution: When approaching upper limits of curve care must be taken because individual well characteristics may differ significantly from these curves.

FIG. 7
EFFECT OF EMULSION ON OIL VISCOSITY

FIG. 7 EFFECT OF EMULSION ON OIL VISCOSITY

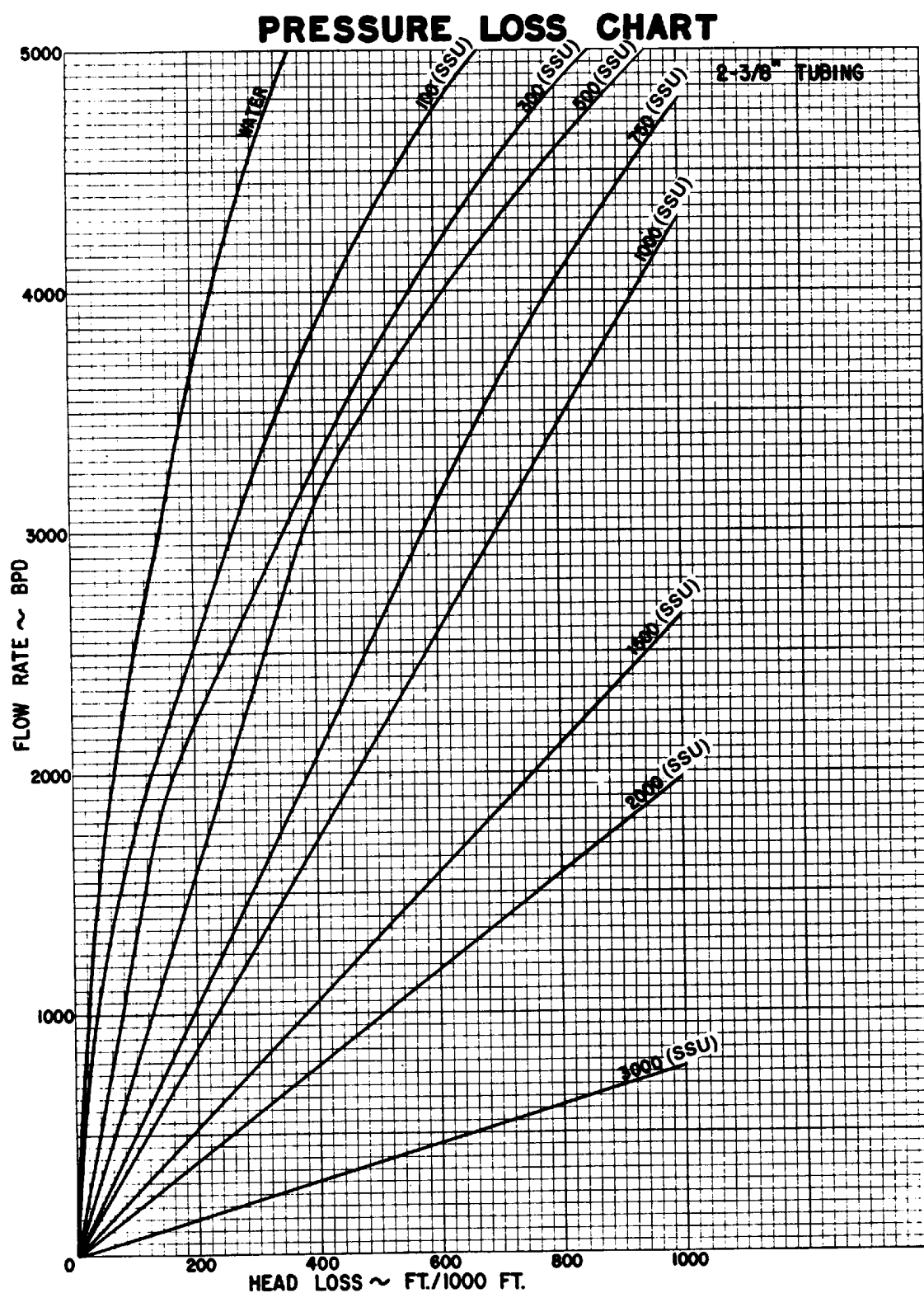


FIG. 8

FIG. 8 PRESSURE LOSS CHART

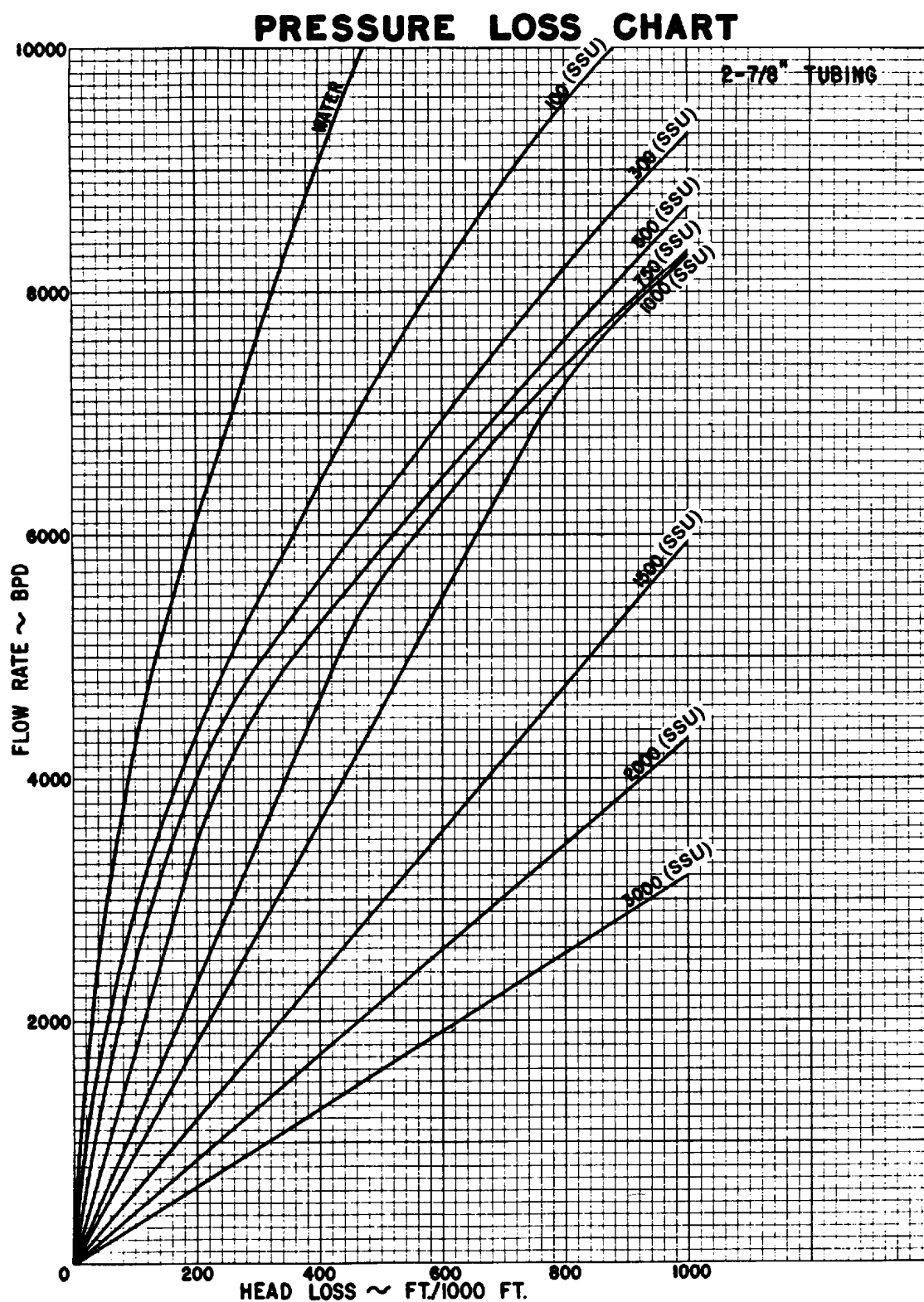


FIG. 9

FIG. 9 PRESSURE LOSS CHART

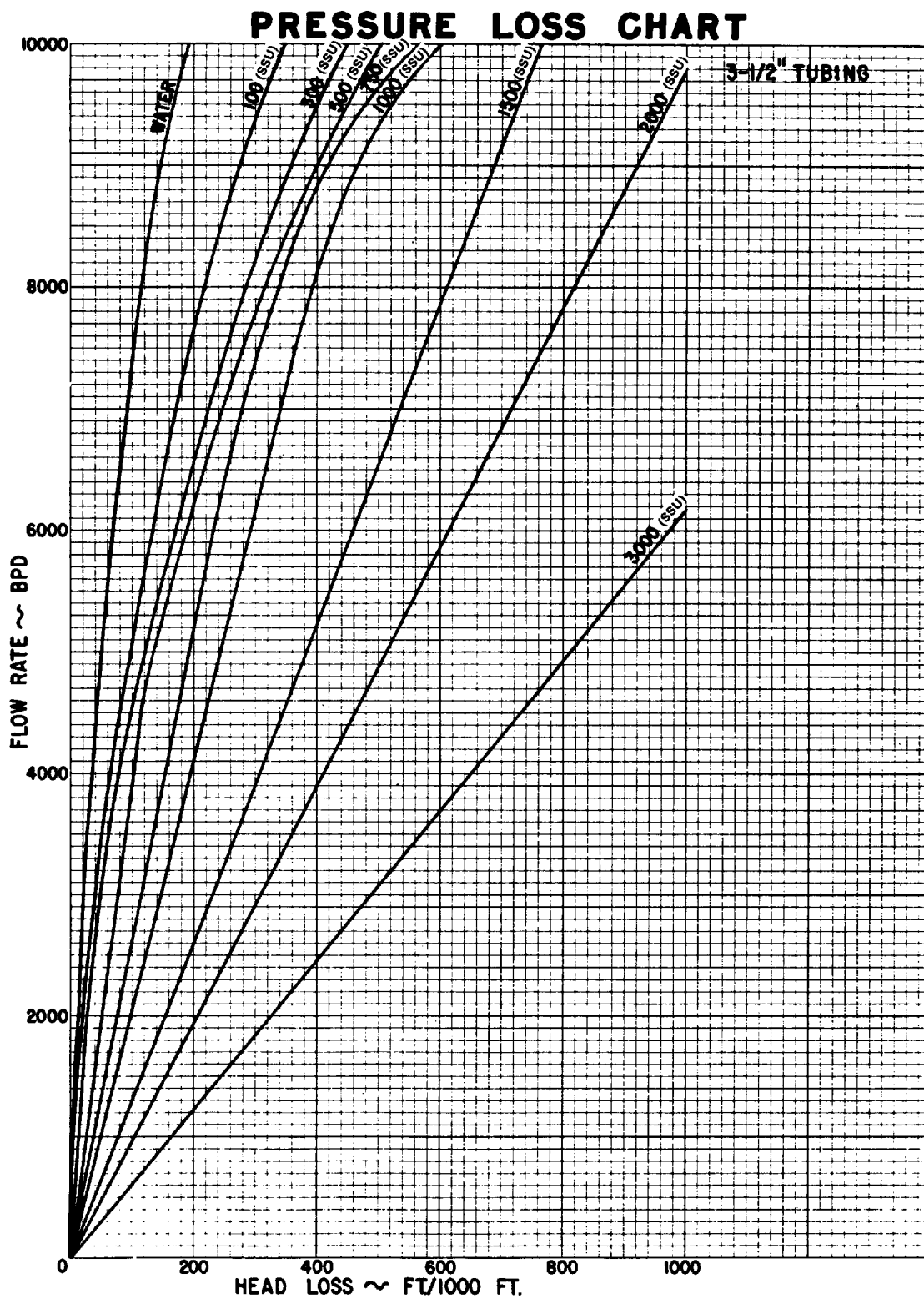


FIG. 10

FIG. 10 PRESSURE LOSS CHART

FIG. 11
CONVERSION CHART
APPROXIMATE CHANGES DUE TO VISCOSITY
(60% MAXIMUM EFFICIENCY PUMPS)

VISCOSITY AT PUMPING TEMPERATURES (SSU)	CAPACITY FACTOR	HEAD FACTOR	HORSEPOWER FACTOR
50	1.00	0.995	1.04
80	0.985	0.985	1.08
100	0.980	0.980	1.11
150	0.960	0.960	1.16
200	0.940	0.940	1.19
300	0.910	0.910	1.24
400	0.880	0.890	1.27
500	0.850	0.860	1.28
600	0.830	0.845	1.30
700	0.805	0.825	1.30
800	0.790	0.810	1.31
900	0.770	0.800	1.32
1000	0.775	0.780	1.33
1500	0.690	0.725	1.36
2000	0.630	0.675	1.31

FIG. 11 CONVERSION CHART APPROXIMATE CHANGES DUE TO VISCOSITY
(60% MAXIMUM EFFICIENCY PUMPS)

VOLTAGE DROP IN CABLE

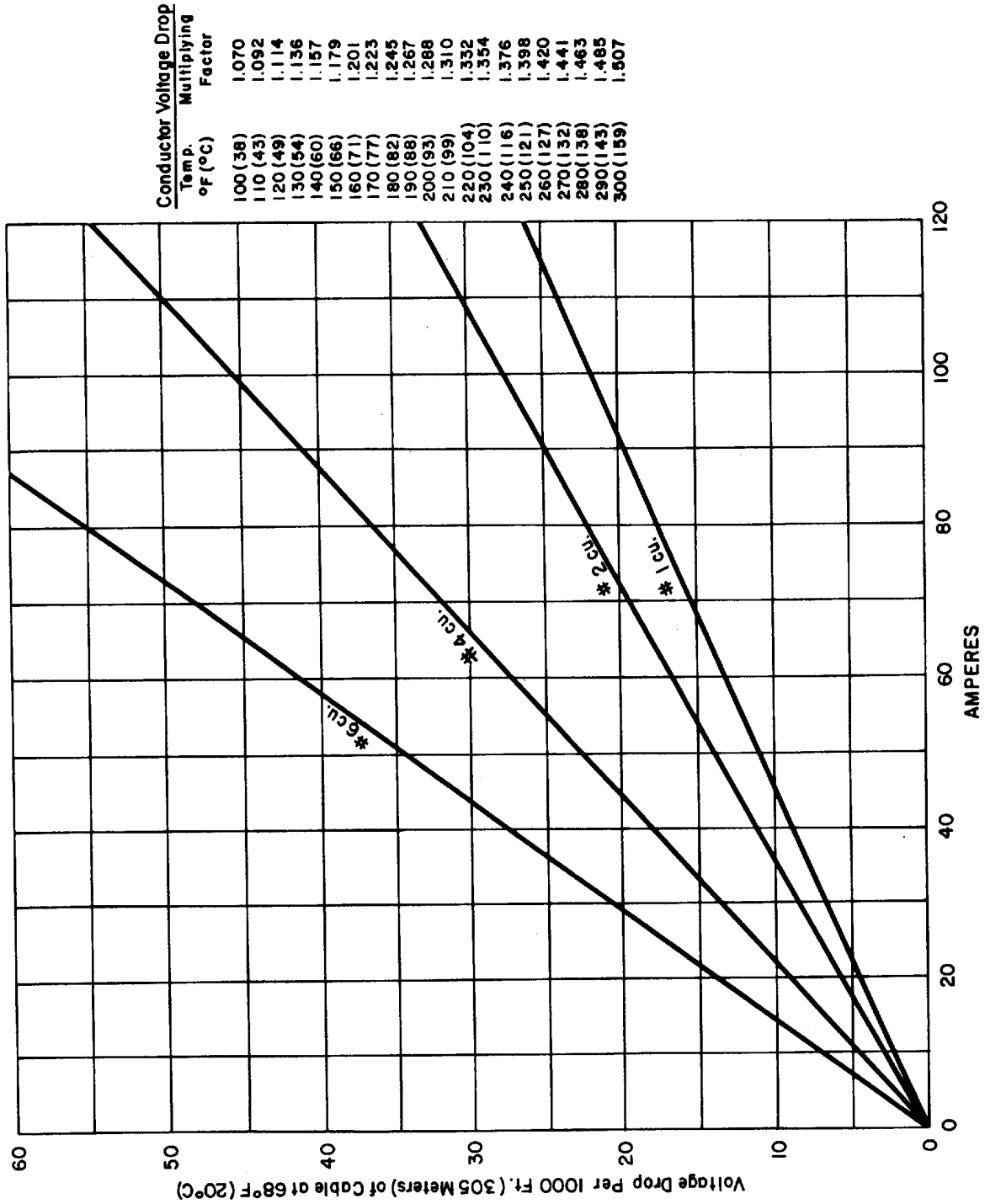
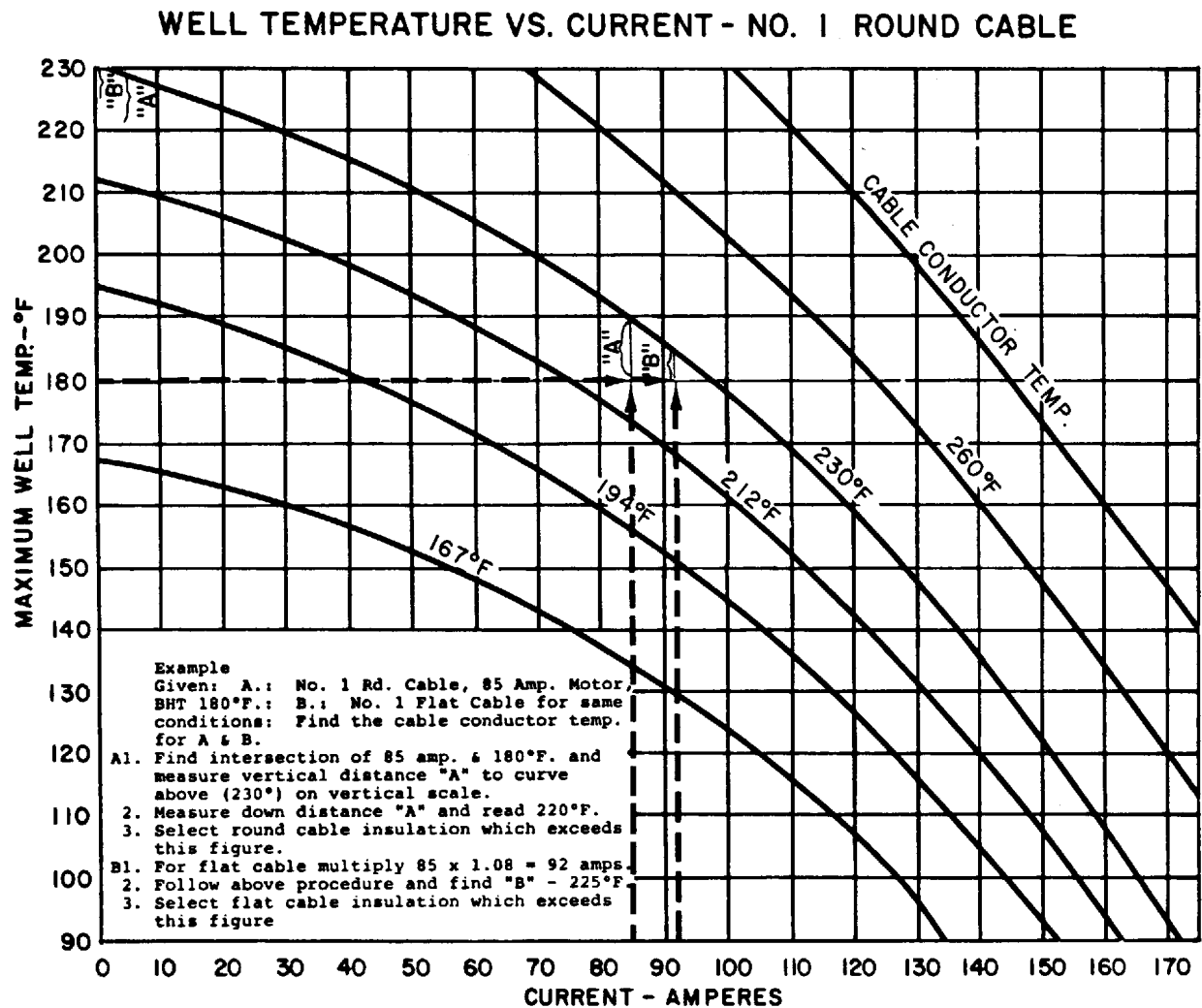


FIG. 12 VOLTAGE DROP IN CABLE



**FIG. 13
WELL TEMPERATURE VS. CURRENT**

FIG. 13 WELL TEMPERATURE VS. CURRENT

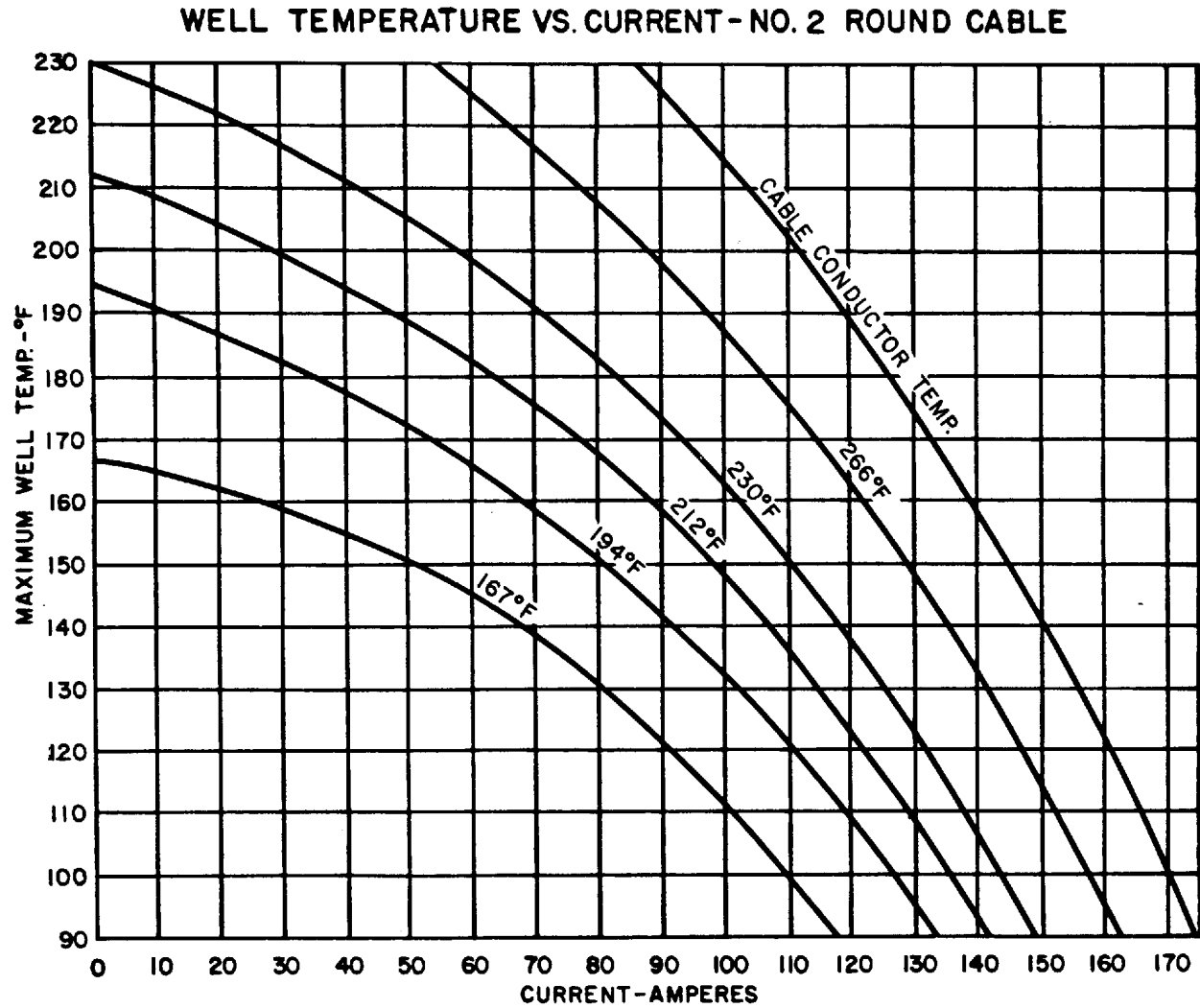


FIG. 14
WELL TEMPERATURE VS. CURRENT
NO. 2 ROUND CABLE

FIG. 14 WELL TEMPERATURE VS. CURRENT NO. 2 ROUND CABLE

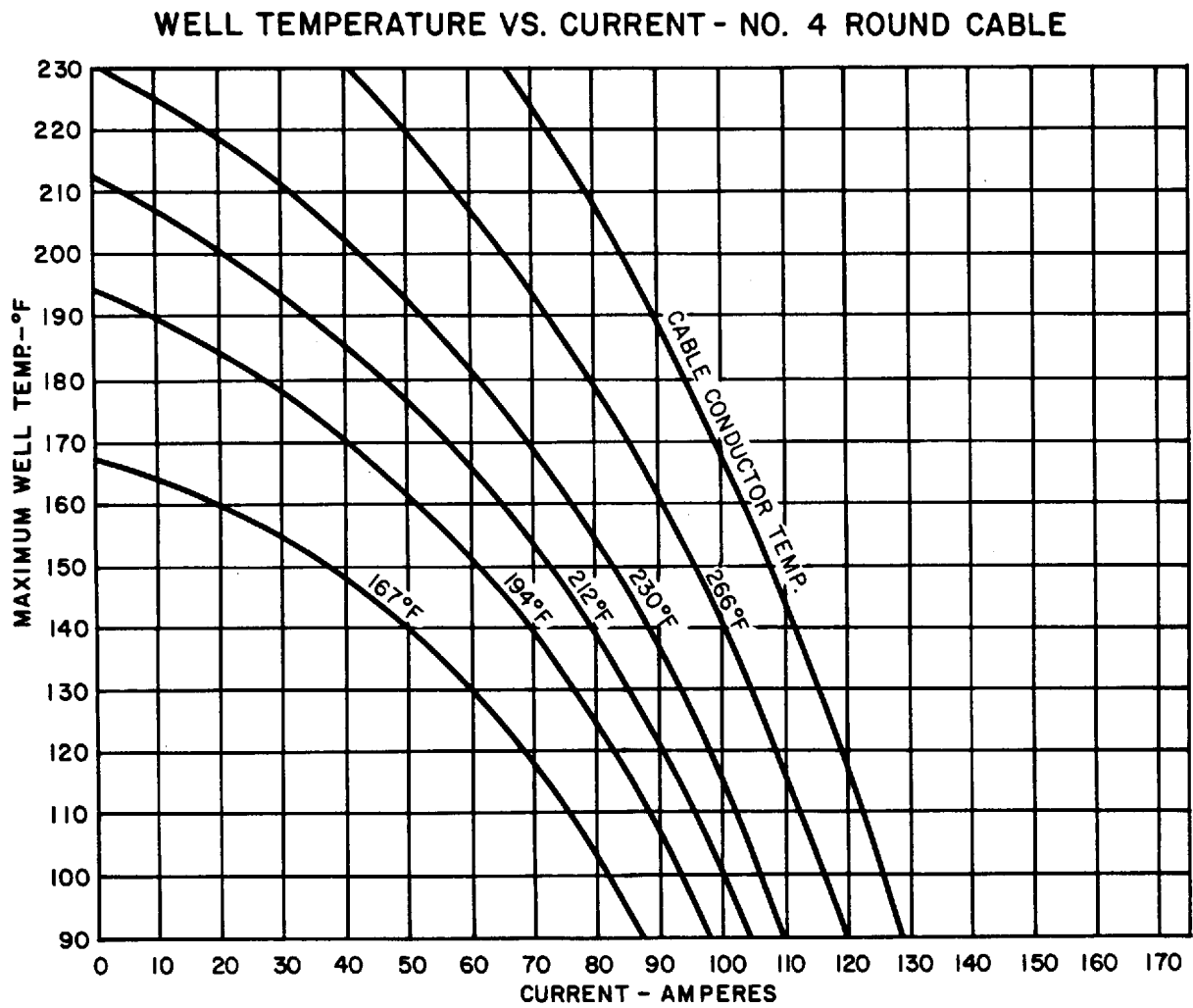


FIG. 15
WELL TEMPERATURE VS. CURRENT
NO. 4 ROUND CABLE

FIG. 15 WELL TEMPERATURE VS. CURRENT NO. 4 ROUND CABLE

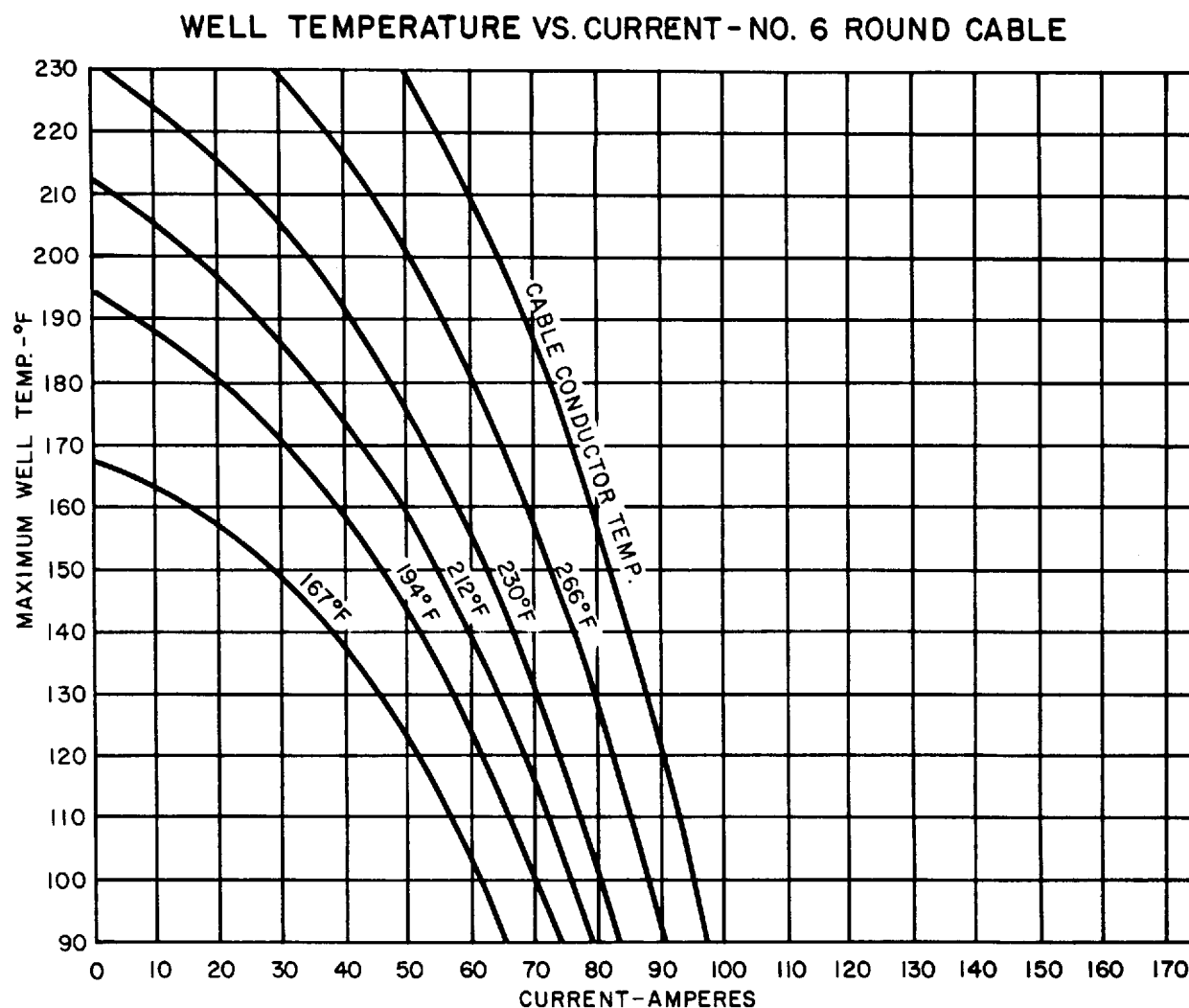


FIG. 16
WELL TEMPERATURE VS. CURRENT
NO. 6 ROUND CABLE

FIG. 16 WELL TEMPERATURE VS. CURRENT NO. 6 ROUND CABLE

PROPERTIES OF NATURAL HYDROCARBON MIXTURES OF GAS AND LIQUID BUBBLE POINT PRESSURE

EXAMPLE

Required:

Bubble point pressure at 200° F of a liquid having a gas-oil ratio of 350 CFB, a gas gravity of 0.75 and a tank oil gravity of 30° API.

Procedure:

Starting at the left side of the chart, proceed horizontally along the 350 CFB line to a gas gravity of 0.75. From this point drop vertically to the 30° API line. Proceed horizontally from the tank oil gravity seals to the 200° F line. The required pressure is found to be 1930 PSIA.

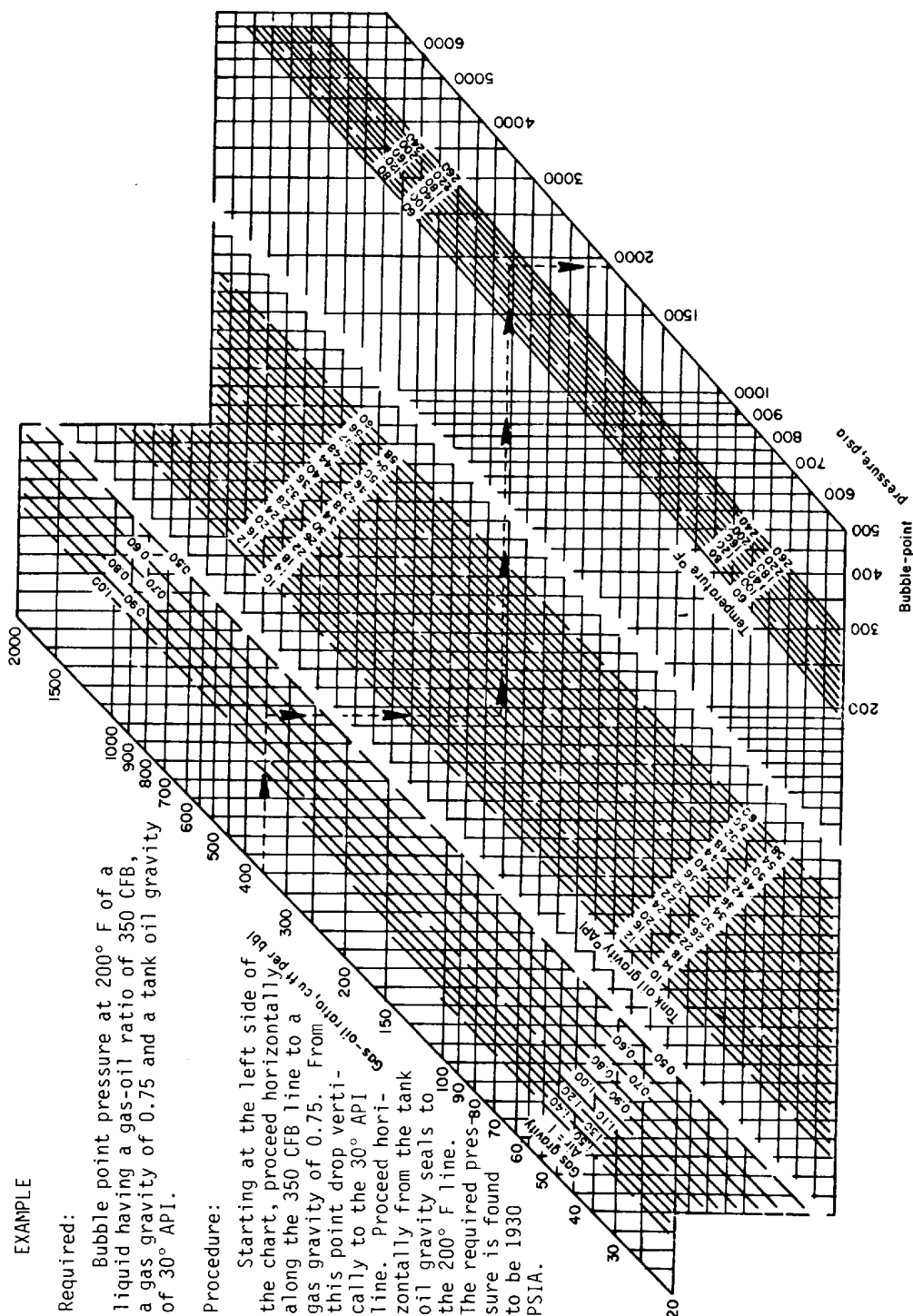


FIG. 17
CHART FOR CALCULATING BUBBLE-POINT PRESSURE OR SOLUBILITY

FIG. 17 CHART FOR CALCULATING BUBBLE-POINT PRESSURE OR SOLUBILITY

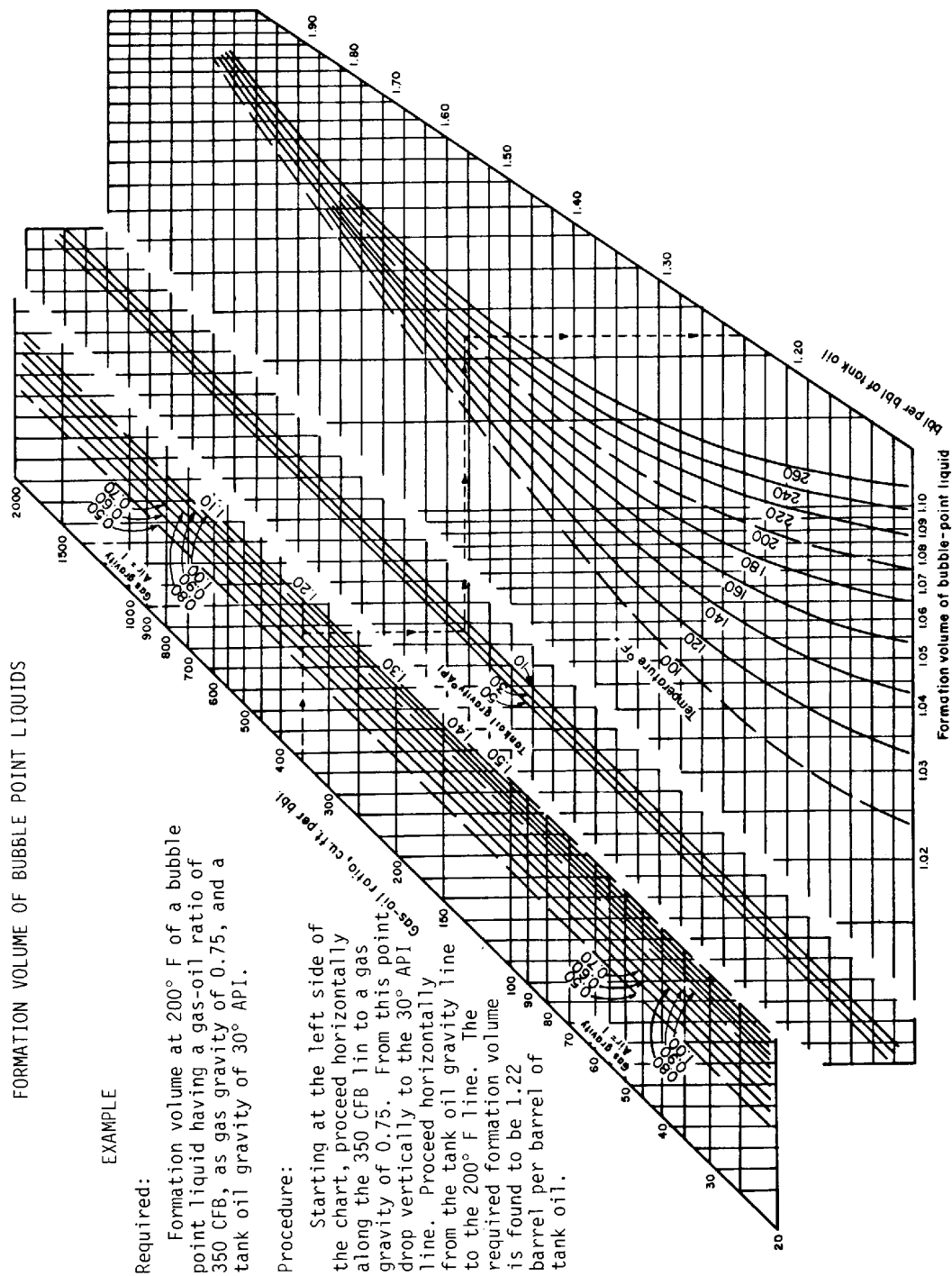


FIG. 18
CHART FOR CALCULATING FORMATION VOLUME OF BUBBLE-POINT LIQUIDS

FIG. 18 CHART FOR CALCULATING FORMATION VOLUME OF BUBBLE-POINT LIQUIDS

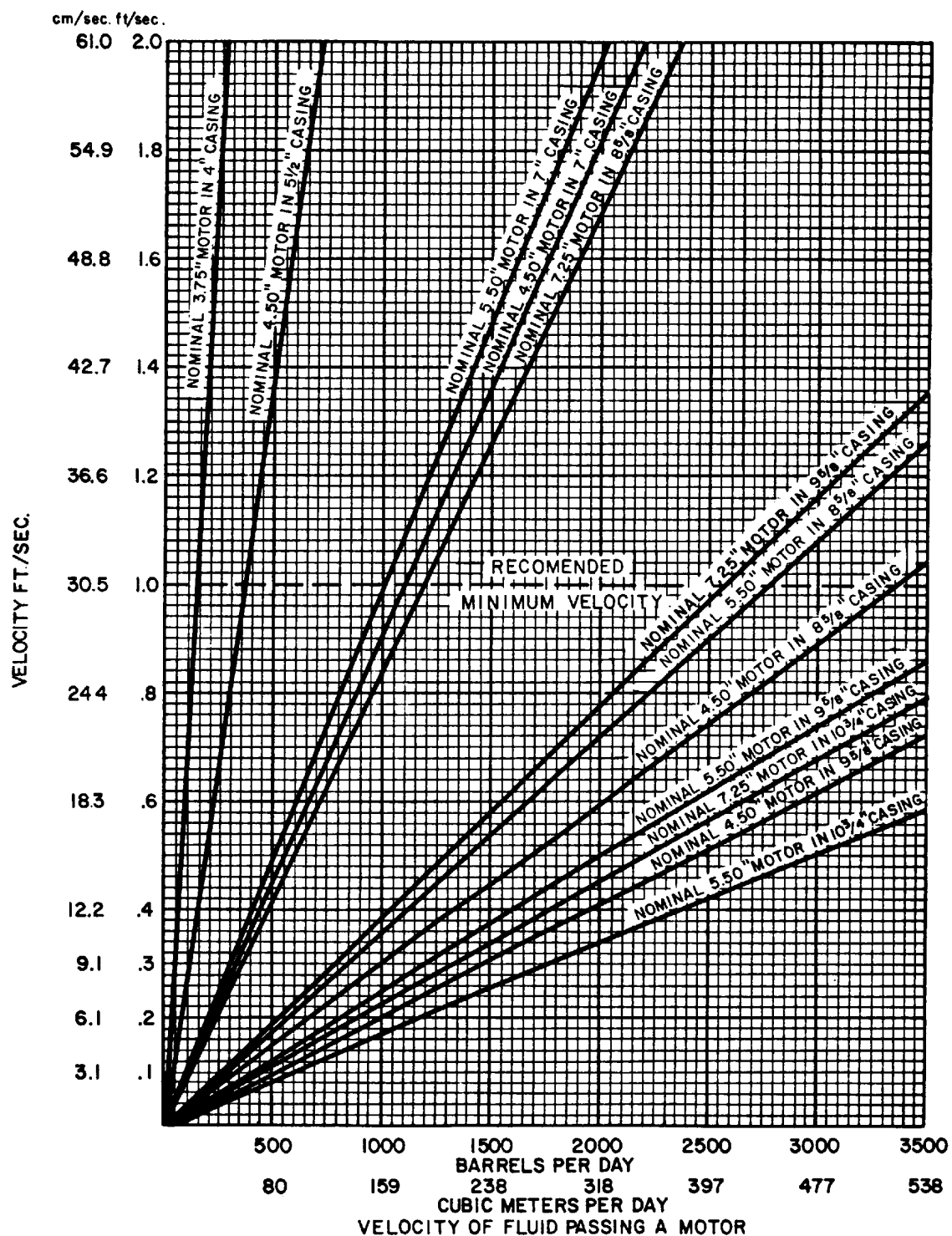


FIG. 19
VELOCITY OF FLUID PASSING A MOTOR

FIG. 19 VELOCITY OF FLUID PASSING A MOTOR

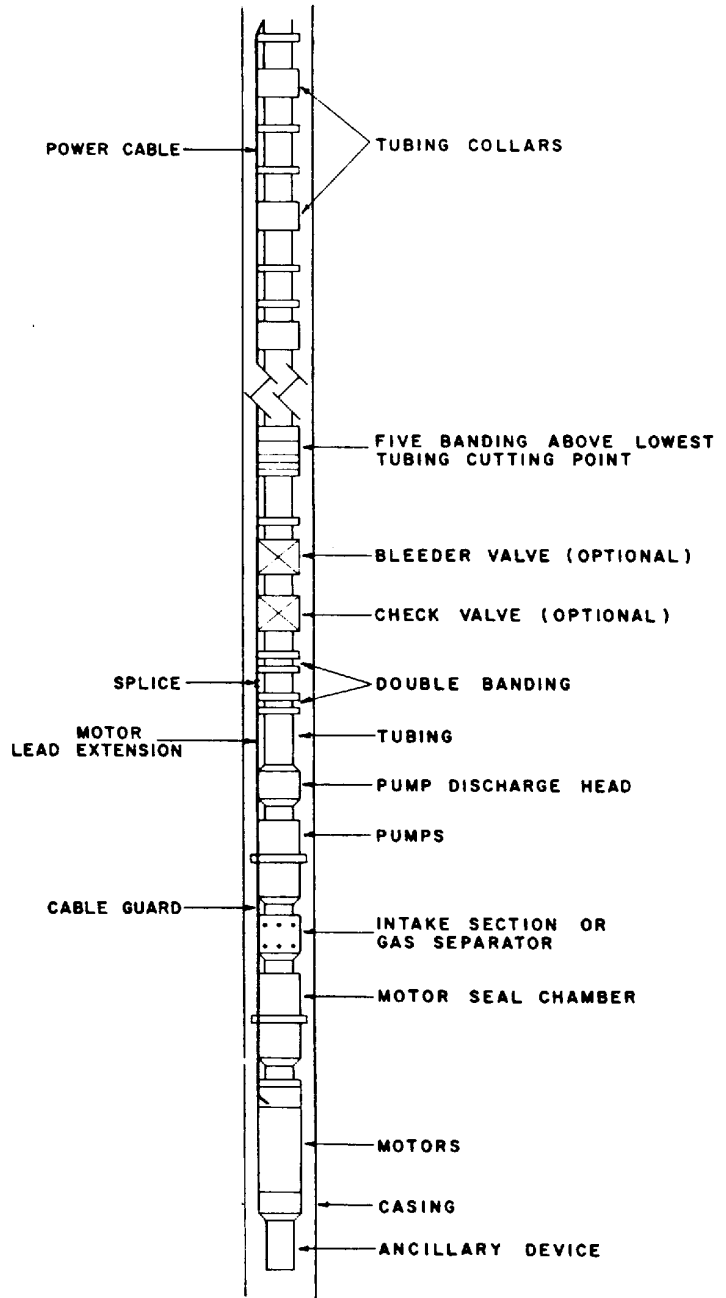
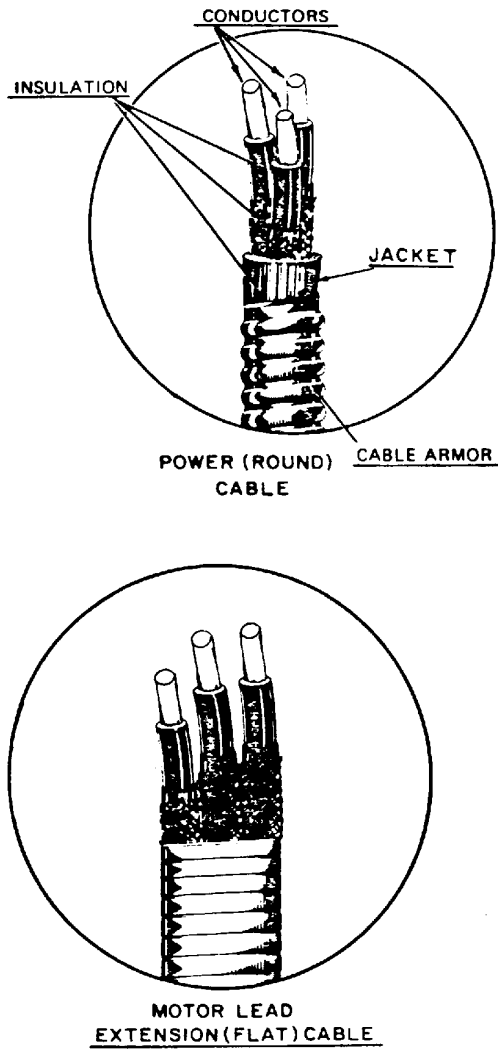


FIG. 20
SUBSURFACE EQUIPMENT

FIG. 20 SUBSURFACE EQUIPMENT

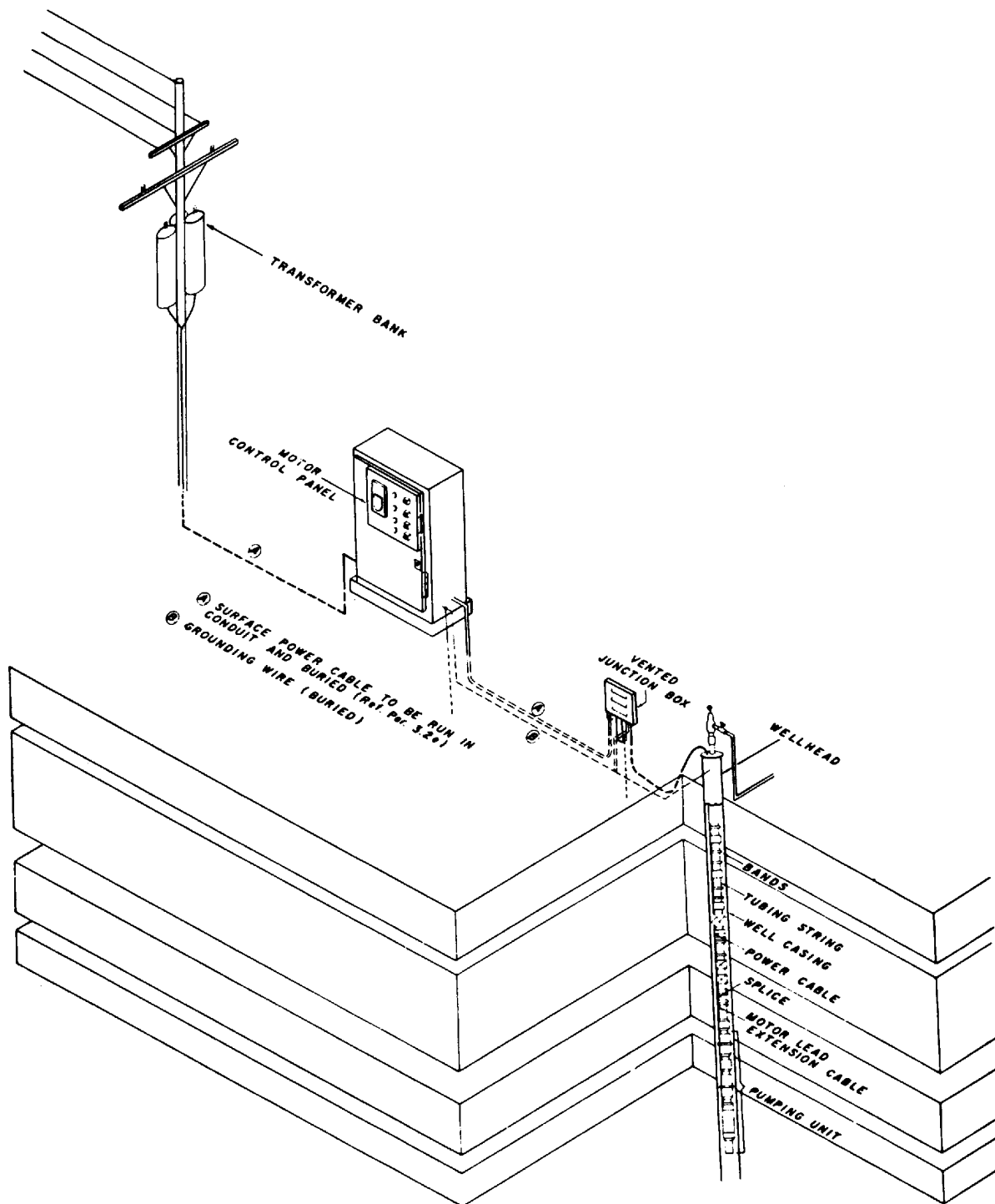


FIG. 21
SURFACE EQUIPMENT

FIG. 21 SURFACE EQUIPMENT

WELL DATA SHEET

A. WELL DATA

1. Well No. & Field Name				Date
2. Well Location (County, State, Other)				
3. Installation				New () or Redesign ()
4. Producing Formations				
5. Formation Type (Sandstone, Limestone, Other)				
6.*API Casing	In. O.D.:	#/Ft.	Ft. TO	Ft.
7.*Liner	In. O.D.:	#/Ft.	Ft. TO	Ft.
8.*Other Limiting Dimensions		In. I.D.	Ft. TO	Ft.
9.*Open Hole			Ft. TO	Ft.
10.*Total Depth		Ft. Plugged Back Depth		Ft.
11.*Perforated Intervals			Ft. TO	Ft.
			Ft. TO	Ft.
12.*Wellhead Type & Size				
13.*API Tubing	In. O.D.	Threads	Grade	

B. RESERVOIR DATA (FROM TEST & PRODUCTION DATA)

1.*Present Production	BOPD	BWPD	BFPD	Pumping ()	Flowing ()
2. Bottom Hole Static Pressure		PSIG AT:		Ft.	
3. Bottom Hole Producing Pressure		PSIG AT:		BFPD	
		PSIG AT:		BFPD	
4. GOR		S.C.F./S.T.B.			
5.*Water Cut		%			
6. Oil API Gravity		°API °			
7. Water Specific Gravity					
8. Gas Specific Gravity					
9. Bottom Hole Temperature		°F			
10. Oil Viscosity (1)		CP.OR		SSU At	°F
(2)		CP.OR		SSU At	°F
11. PVT Data		Solution GOR		FVF	PSIG
		Solution GOR		FVF	PSIG
		Solution GOR		FVF	PSIG
		Solution GOR		FVF	PSIG
12. Bubble Point Pressure		PSIG			

C. SIZING SPECIFICATIONS

1.*Desired Production	BFPD		BOPD		BWPD	
2. Desired Pump (Intake) Vertical Setting Depth		Ft.				
3. Desired Pump (Intake) Pressure		PSIG				
4. Required Wellhead Pressure		PSIG				
5. Gas Through Pump		% (Of Total Gas)				
6. Primary Voltage Available		Volts		Hertz		
7. Casing Pressure		PSIG				
8. Special Problems:	Sand ()	Scale ()	Corrosion ()	Paraffin ()	H ₂ S ()	CO ₂ ()

REMARKS:

BY: _____

DATE: _____

*Indicates Data That Must Be Provided

**FIG. 22
ELECTRIC SUBMERSIBLE PUMP
WELL DATA SHEET**

FIG. 22 ELECTRIC SUBMERSIBLE PUMP WELL DATA SHEET

WELL DATA SHEET

A. WELL DATA

1. Well No. & Field Name		EXAMPLE PROBLEM NO. 1		Date	10-25-82
2. Well Location (County, State, Other) OIL & WATER — NO GAS					
3. Installation				New (X)	or Redesign ()
4. Producing Formations					
5. Formation Type (Sandstone, Limestone, Other)					
6.*API Casing 5½	In. O.D.: 15.5 & 17	#/Ft. 10,000	Ft. TO	Ft.	
7.*Liner NONE	In. O.D.:	#/Ft.	Ft. TO	Ft.	
8.*Other Limiting Dimensions	NONE	In. I.D.	Ft. TO	Ft.	
9.*Open Hole	NONE		Ft. TO	Ft.	
10.*Total Depth 10,000	Ft. Plugged Back Depth	9960		Ft.	
11.*Perforated Intervals			9800 Ft. TO 9840	Ft.	
			9880 Ft. TO 9900	Ft.	
12.*Wellhead Type & Size HHS 2%					
13.*API Tubing	2% In. O.D.	8EU Threads	J-55	Grade	

B. RESERVOIR DATA (FROM TEST & PRODUCTION DATA)

1.*Present Production	50 BOPD	200 BWPD	250 BFPD	Pumping (X)	Flowing ()
2. Bottom Hole Static Pressure	4000 PSIG AT:		9850 Ft.		
3. Bottom Hole Producing Pressure	3300 PSIG AT:		250 BFPD		
		PSIG AT:	BFPD		
4. GOR	NIL	S.C.F./S.T.B.			
5.*Water Cut	80.0%				
6. Oil API Gravity	37.0 °API	0.840 SP.GR.°			
7. Water Specific Gravity	1.060				
8. Gas Specific Gravity	N.A.				
9. Bottom Hole Temperature	190 °F @ PUMP SETTING DEPTH				
10. Oil Viscosity (1)		CP.OR	40 SSU At	195 °F	
(2)		CP.OR	SSU At	°F	
11. PVT Data	NONE	Solution GOR	FVF	PSIG	
		Solution GOR	FVF	PSIG	
		Solution GOR	FVF	PSIG	
		Solution GOR	FVF	PSIG	
12. Bubble Point Pressure	500 PSIG				

C. SIZING SPECIFICATIONS

1.*Desired Production	1100 BFPD	220 BOPD	880 BWPD
2. Desired Pump (Intake) Vertical Setting Depth	9000 Ft.		
3. Desired Pump (Intake) Pressure	500 PSIG		
4. Required Wellhead Pressure	100 PSIG		
5. Gas Through Pump	0% (Of Total Gas)		
6. Primary Voltage Available	14,400/24,900	Volts	60 Hertz
7. Casing Pressure	0 PSIG		
8. Special Problems: NONE	Sand ()	Scale ()	Corrosion () Paraffin () H ₂ S () CO ₂ ()

REMARKS:

Operator desires stainless steel bands, three single phase transformers,
and no check valve or drain valve

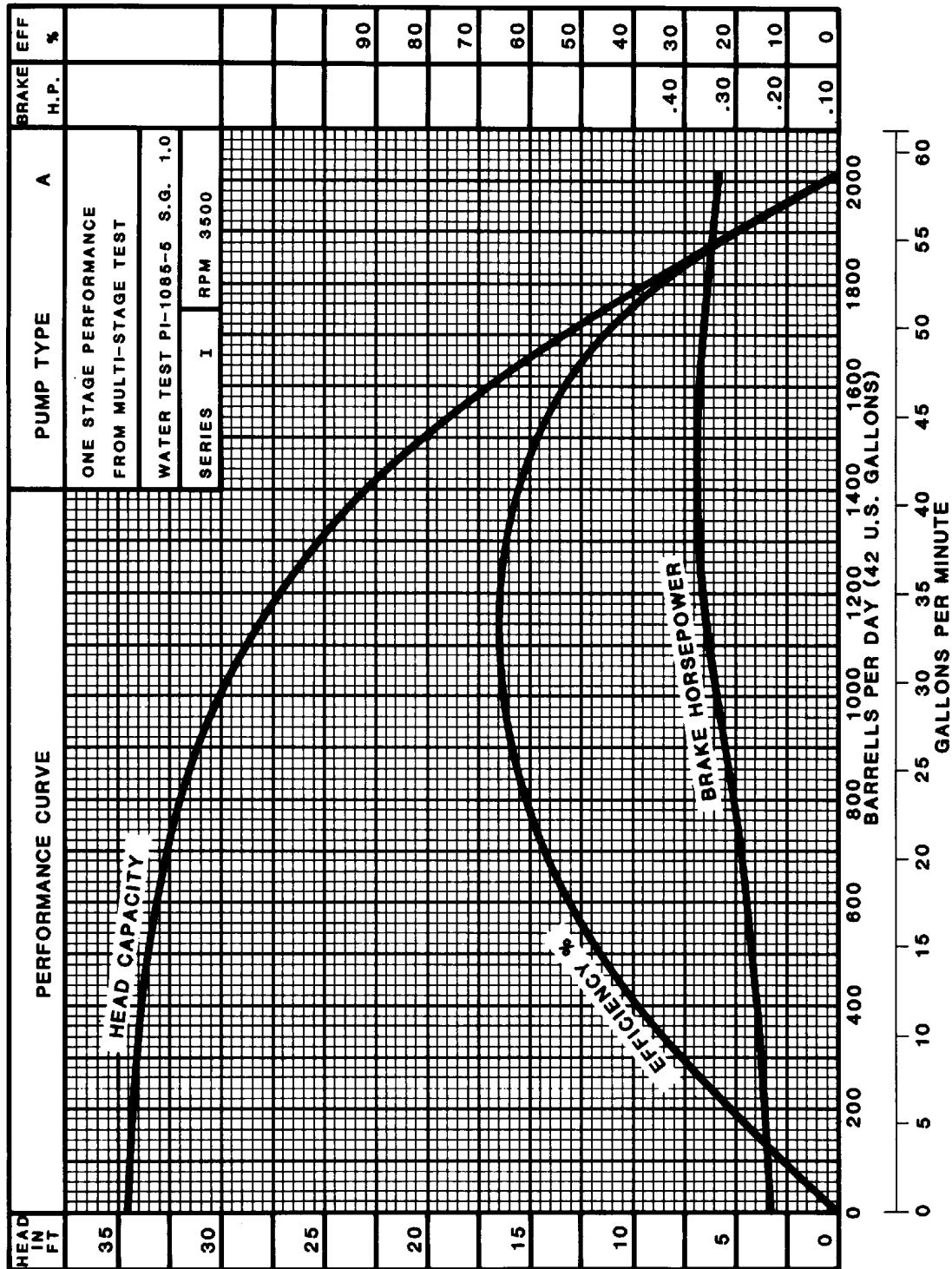
BY: A. B. Jones

DATE: 10-25-82

*Indicates Data That Must Be Provided

EXHIBIT 1.1 ELECTRIC SUBMERSIBLE PUMP WELL DATA SHEET EXAMPLE PROBLEM NO. 1

EXHIBIT 1.1 ELECTRIC SUBMERSIBLE PUMP WELL DATA SHEET EXAMPLE PROBLEM NO. 1



PUMP TYPE A FOR 5 1/2" O.D. CASING

**EXHIBIT 1.2
PUMP TYPE A FOR 5 1/2" O.D. CASING
PUMP TYPE A**

EXHIBIT 1.2 PUMP TYPE A FOR 5½" O.D. CASING PUMP TYPE A

SERIES I, PUMP TYPE A
(FOR 5 1/2" O.D. AND LARGER CASINGS)
SPECIFICATIONS
PUMP DISCHARGE 2 3/8" O.D. EUE 8 ROUND THREAD

<u>HOUSING NO.</u>	<u>SHIPPING LENGTH (FEET)</u>	<u>WEIGHT (POUNDS)</u>	<u>NO. OF STAGES</u>
2	5.2	145	26
2	5.2	155	34
3	6.7	188	---
3	6.7	198	51
4	8.2	231	59
4	8.2	241	68
5	9.7	274	---
5	9.7	284	79
5	9.7	294	85
6	11.2	327	101
7	12.7	370	118
8	14.2	413	135
9	15.7	446	152
10	17.2	489	169
11	18.7	532	186
12	20.2	575	203
13	21.7	618	220
14	23.2	661	237
15	24.7	702	254
16T	27.7	820	271
17T	29.2	863	288
18T	30.7	906	305
19T	32.2	949	322
20T	33.7	992	339
21T	35.2	1035	355
22T	36.7	1078	372
23T	38.2	1121	389
24T	39.7	1164	406
25T	41.2	1207	423
26T	42.7	1250	440
27T	44.2	1297	457
28T	45.7	1340	474
29T	47.2	1383	491
30T	48.7	1426	508

1. Housing numbers ending with "T" are 2-piece pumps.
2. Above pumps fit I Series seal section.

EXHIBIT 1.3
PUMP TYPE A

EXHIBIT 1.3 PUMP TYPE A

MOTORS
(60 CYCLE, 3500 RPM)
RATING, SPECIFICATIONS

SERIES I FOR 5 1/2" O.D. AND LARGER CASING

<u>H.P.</u>	<u>VOLTS</u>	<u>AMPS</u>	<u>LENGTH</u>	<u>WEIGHT, LBS.</u>
15	430	22	5.8'	263
22½	430	33	7.5'	350
	700	21		
30	430	44	9.2'	425
	720	27		
37½	450	53	10.9'	512
	720	33		
	890	28		
45	750	38	12.6'	595
	1080	27		
52½	750	44	14.3'	687
	1130	30		
60	720	53	16.0'	763
	1150	33		
75	1250	38	19.3'	935
90	1290	45	22.7'	1095
	2150	27		
105	1260	53	26.1'	1265
	2270	30		
120	1150	67	29.5'	1430
	2280	34		

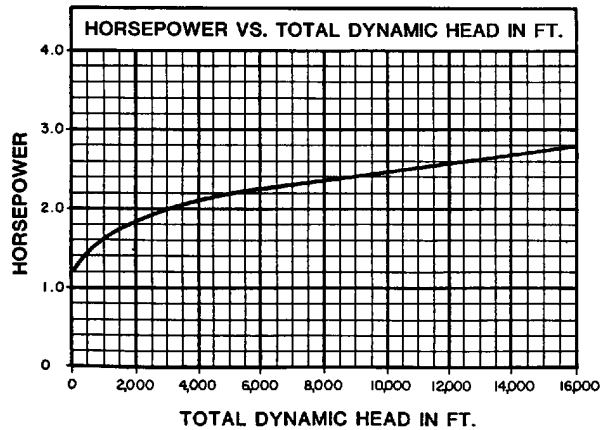
SERIES I MOTORS LISTED BELOW ARE TANDEM (Two Piece) CONSTRUCTION

<u>H.P.</u>	<u>VOLTS</u>	<u>AMPS</u>	<u>LENGTH</u>	<u>WEIGHT, LBS.</u>
150	2130	46	19.5' & 19.5'	1875
180	2120	55	22.9' & 22.9'	2195
210	2280	59	26.3' & 26.3'	2535
240	2250	68	29.5' & 29.5'	2860

EXHIBIT 1.4
MOTORS
(60 Cycle, 3500 RPM)
RATING, SPECIFICATIONS

EXHIBIT 1.4 MOTORS (60 Cycle, 3500 RPM) RATING, SPECIFICATIONS

MOTOR SEAL CHAMBER HP CURVE



MODEL	MINIMUM CASING SIZE (INCHES)	SHIPPING LENGTH (FEET)	WEIGHT (POUNDS)
1	5½	4.5'	110
2	5½	5.5'	130
3	7	4.5'	175
4	7	5.6'	195

NOTE: Model 2 is designed for use with Tandem (150 HP or larger) Motors. Model 4 is designed for use with Tandem (360 HP or larger) Motors.

**EXHIBIT 1.5
MOTOR SEAL CHAMBER**

EXHIBIT 1.5 MOTOR SEAL CHAMBER

NEMA SIZE	RATINGS			DIMENSIONS & WEIGHTS			
	VOLTS	MAX. AMPS	MAX. H.P.	HEIGHT	WIDTH	DEPTH	WEIGHT (POUNDS)
A2	440	50	25	48	28	10½	215
A3	440	100	50	48	28	10½	290
A4	440	150	100	48	28	10½	310
A5	440	300	200	48	28	10½	350
B2	880	50	40	48	28	10½	290
B3	880	100	75	48	28	10½	295
C3	1500	100	150	71	28	20	680
C4	1500	150	225	71	28	20	710
D3	2500	200	700	67	31	38	930
D33	3300	200	700	67	31	38	935

**EXHIBIT 1.6
MOTOR CONTROL PANEL & ACCESSORIES
440 VOLT, 880 VOLT, 1500 VOLT, 2500 VOLT
DESCRIPTION**

EXHIBIT 1.6 MOTOR CONTROL PANEL & ACCESSORIES 440 VOLT, 880 VOLT, 1500 VOLT, 2500 VOLT

DESCRIPTION

SINGLE-PHASE, OIL IMMERSED SELF COOLED TYPE TRANSFORMERS

TRANSFORMERS ARE AVAILABLE IN THE PRIMARY VOLTAGES OF 2400/4160 Y;
7200/12,470 Y; 13,200; 14,400/24,940 Y.

KVA SIZE	SECONDARY VOLTAGE	HEIGHT	WIDTH	DEPTH	WEIGHT
28	625/1750	36"	21"	24"	400#
37.5	625/1750	39"	23"	26"	550#
50	625/1750	45"	23"	26"	670#
75	625/1750	49"	28"	29"	980#
75	1900/2500	49"	28"	29"	980#
100	625/1750	49"	33"	29"	1,125#
100	1900/2500	49"	33"	29"	1,125#

NOTE: Secondary voltages above are those obtained with Delta connected secondary. Taps are provided for numerous intermediate voltages. Additional secondary voltages can be obtained by WYE connecting the secondary.

THREE PHASE AUTO TRANSFORMERS, OIL IMMERSED SELF COOLED.

KVA	PRIMARY VOLTAGE	SECONDARY VOLTAGE	SECONDARY AMPS		DIMENSIONS			
			FROM H.V. TAP SETTING	TO L.V. TAP SETTING	HEIGHT	WIDTH	DEPTH	WEIGHT
50	440/460/	750-1050	27	38	33"	27"	28"	650#
50	480	800-1250	23	36				
75	"	800-1250	35	55	33"	27"	32"	900#
75	"	850-1450	30	51				
100	"	850-1450	40	68	37"	35"	32"	1120#
125	"	850-1450	50	85	37"	35"	34"	1250#
150	"	850-1450	60	102	42"	33"	34"	1499#
200	"	850-1450	80	136	43"	40"	37"	1880#
250	"	850-1450	100	170	43"	40"	37"	1920"

NOTE:

1. Pole hangers not available for 200 & 250 KVA.
2. Voltage steps are 50 V on 750-1050, 75 V on 800-1250 and 100 V on 850-1450 secondary voltages. However, intermediate voltages are available by taking advantage of the three primary taps.

EXHIBIT 1.7 TRANSFORMERS

EXHIBIT 1.7 TRANSFORMERS

DESCRIPTION (TYPE)	CASING SIZE (O.D.)	TUBING SIZE (O.D.)	MAXIMUM RECOMMENDED LOAD (LBS)	WEIGHT (LBS)
A	5 1/2"	2 3/8"	30,000	57
		2 7/8"	30,000	57
	6 5/8", 7"	2 3/8"	30,000	70
		2 7/8"	30,000	70
	5 1/2", 7"	2 3/8"	80,000	147
		2 7/8"	80,000	147
B	7"	3 1/2"	80,000	146
	8 5/8"	2 3/8" - 3 1/2"	100,000	197
		4 1/2"	100,000	191
		5 1/2"	100,000	191
	9 5/8", 10 3/4"	2 3/8"	100,000	267
		2 7/8"	100,000	267
		3 1/2"	100,000	267
		4 1/2"	100,000	264
		5 1/2"	100,000	264
		6 5/8"	100,000	264

**EXHIBIT 1.8
TUBING HEADS
RATING, SPECIFICATIONS**

EXHIBIT 1.8 TUBING HEADS RATING, SPECIFICATIONS

3KV ROUND CABLE W/I-WIRE							
CONDUCTOR SIZE	CONDUCTOR MATERIAL	CONDUCTOR CONST.	INSULATION	JACKET	ARMOR	PART NUMBER	WT. LBS PER FOOT
#1	Copper	Stranded	PPE	Nitrile	Galv. Stl.	A2638437	1.88
#2	Copper	Stranded	PPE	Nitrile	Galv. Stl.	A2638436	1.62
#4	Copper	Solid	PPE	Nitrile	Galv. Stl.	A2638434	1.15
#6	Copper	Solid	PPE	Nitrile	Galv. Stl.	A2638435	.88
#4	Copper	Solid	PPE	*	*	A2641415	.52
#6	Copper	Solid	PPE	*	*	A2641416	.44

3KV ROUND CABLE WITHOUT I-WIRE							
#4	Copper	Solid	PPE	Nitrile	Galv. Stl.	A2642201	1.15
#6	Copper	Solid	PPE	Nitrile	Galv. Stl.	A2642202	.88

* High Density Polyethylene

3KV FLAT CABLE							
#1	Copper	Solid	PPE	Nitrile	Galv. Stl.	A2640131	1.72
#1	Copper	Solid	EPR	**	Galv. Stl.	A2640132	2.60
#2	Copper	Solid	PPE	Nitrile	Galv. Stl.	A2641234	1.48
#2	Copper	Solid	EPR	**	Galv. Stl.	A2641235	2.26
#4	Copper	Solid	PPE	Nitrile	Galv. Stl.	A2638789	1.06
#4	Copper	Solid	EPR	**	Galv. Stl.	A2638438	1.69

** Lead w/Nylon Braid

**EXHIBIT 1.9
CABLE SPECIFICATIONS**

EXHIBIT 1.9 CABLE SPECIFICATIONS

* MAXIMUM ARMORED CABLE SIZE RECOMMENDED WITH VARIOUS TUBING SIZES (O.D.)														
EQUIPMENT SERIES	CASING O.D. (INCHES)	CASING WEIGHT (#/FT)	NOMINAL I.D.	API EXTERNAL UPSET					API NON-UPSET					
				NOMINAL O.D. COUPLING O.D.	2 3/8	2 7/8	3 1/2	2 3/8	2 7/8	3 1/2	4 1/2	5 1/2		
I	5 1/2	14	5.012		1	6			1		4			
		15.5	4.950		1	6			1		4			
		17	4.892		1	6			1		6			
		20	4.778		1				1		6			
II	6 5/8	17		1	1		6	1	1	1	4			
		20	6.049	1	1		4	1	1	1	4			
II	7	17	6.538		1	1		1	1	1	1			
		20	6.456		1	1		1	1	1	1			
		23	6.366		1	1		1	1	1	1			
		26	6.276		1	1		1	1	1	1			
		29	6.184		1	1		1	1	1	1			
		32	6.094		1	1		2	1	1	1			
II	8 5/8	49	7.511		1	1		1	1	1	4	1	1	
*Contact Factory if larger size cable is required. Parallel cable of various dimensions is available on special order.														
MAXIMUM DIAMETER OF UNITS (INCHES)														
SERIES	MOTOR	PUMP	MOTOR SEAL CHAMBER	FLAT CABLE	CABLE BANDS	MAX. DIA. WITH GUARDS	MAX. DIA. LESS CABLE GUARDS							
I	4.500	4.00	4.00	.450	.030	4.770	4.730							
II	5.625	5.375	5.375	.485	.030	6.055	6.015							
CABLE INFORMATION														
CABLE SIZE			WEIGHT LBS/FT		DIAMETER				CIRCULAR MILS PER CONDUCTOR					
			ARMORED	UNARMORED	ARMORED	UNARMORED	ARMORED	UNARMORED						
1			1.75	--			1.40	--	83,400					
2			1.50	--			1.32	--	66,200					
4			1.10	.64			1.14	.92	41,640					
6			.82	.46			1.00	.76	26,250					
8			.65	.29			.94	.71	16,500					

EXHIBIT 1.10
EQUIPMENT COMBINATIONS IN VARIOUS CASING SIZES

EXHIBIT 1.10 EQUIPMENT COMBINATIONS IN VARIOUS CASING SIZES

FLAT CABLE EXTENSIONS

MOTOR SERIES	ARMOR	WIRE SIZE	TYPE	WIDTH	LENGTHS AVAILABLE					
					25'	40'	50'	60'	75'	100'
I	Bronze	6	3 KV Hi-Duty	.450	X	X		X	X	X
I	Monel	6	3 KV Hi-Duty	.450	X	X		X	X	X
II	Bronze	5	3 KV Hi-Duty	.485	X	X	X			
II	Monel	5	3 KV Hi-Duty	.485	X	X	X			
II	Monel	4	3 KV Hi-Duty	.527	X				X	

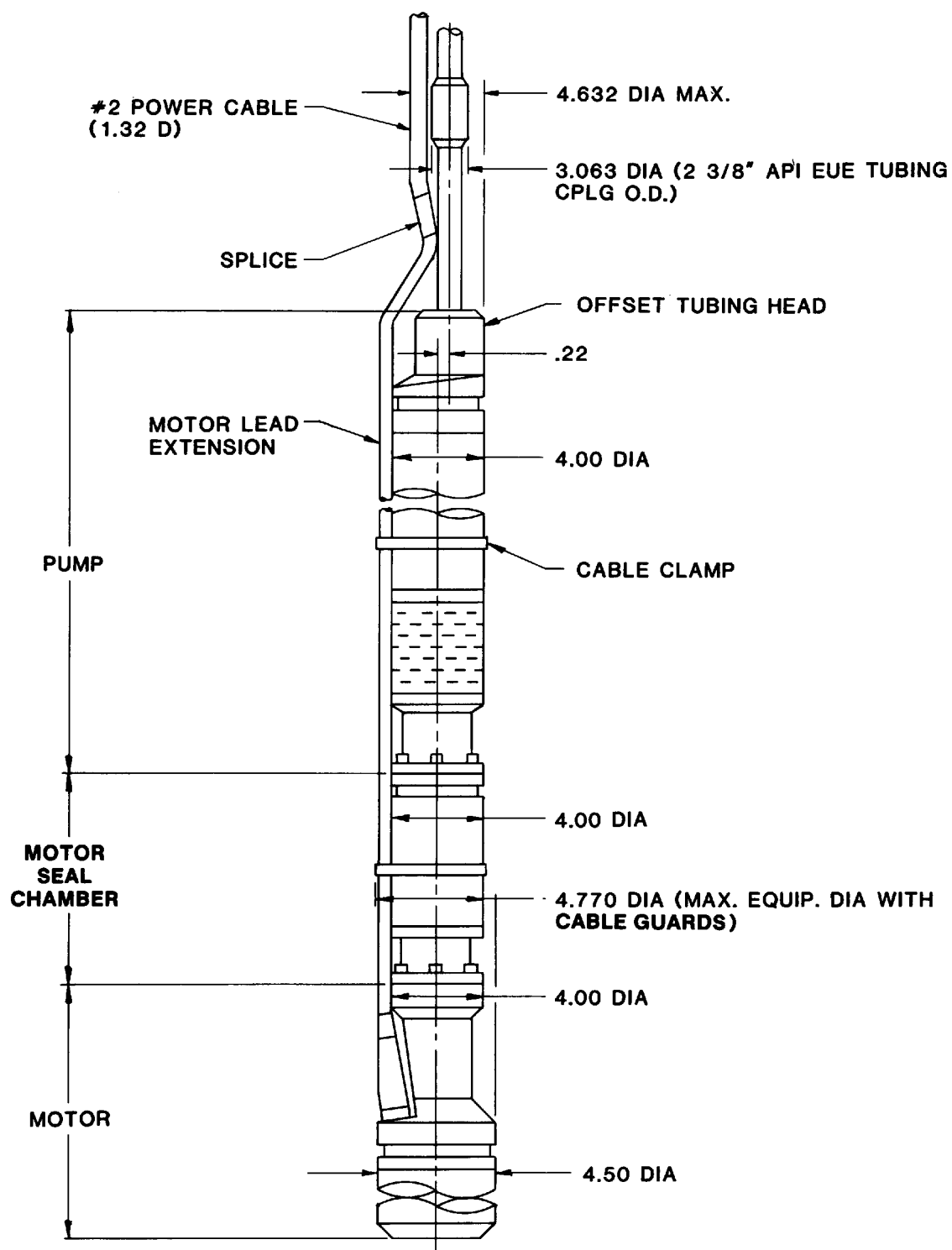
SURFACE CABLE

Surface cables are required to connect power line to motor control panel or motor control panel to submersible cable. Units include connectors on each end.

WEIGHT DIA SIZE (LBS/FT)(INCHES)			3 KV SURFACE CABLE			
			NEW		USED	
			50 FOOT UNIT	PRICE PER ADDITIONAL FOOT	50 FOOT UNIT	PRICE PER ADDITIONAL FOOT
1	1.75	1.40	X	X	X	X
2	1.50	1.32	X	X	X	X
4	1.10	1.14	X	X	X	X
6	0.82	1.00	X	X	X	X

EXHIBIT 1.11 FLAT CABLE AND SURFACE CABLE SPECIFICATIONS

EXHIBIT 1.11 FLAT CABLE AND SURFACE CABLE SPECIFICATIONS



**EXHIBIT 1.12
DOWNHOLE EQUIPMENT WITH
OFFSET TUBING HEAD**

EXHIBIT 1.12 DOWNHOLE EQUIPMENT WITH OFFSET TUBING HEAD

A. WELL DATA

1. Well No. & Field Name				EXAMPLE PROBLEM NO. 2		Date
2. Well Location (County, State, Other)				Oil & Water w/gas		
3. Installation				New (X)	or	Redesign ()
4. Producing Formations				Devonian		
5. Formation Type (Sandstone, Limestone, Other)				Dolomite Limestone		
6.*API Casing 5.5		In. O.D.: 15.5 & 17	#/Ft. Surface	Ft. TO	10,000 Ft.	
7.*Liner NONE		In. O.D.:	#/Ft.	Ft. TO	Ft.	
8.*Other Limiting Dimensions NONE		In. I.D.		Ft. TO	Ft.	
9.*Open Hole NONE				Ft. TO	Ft.	
10.*Total Depth		10,000 Ft. Plugged Back Depth			9960 Ft.	
11.*Perforated Intervals				9800 Ft. TO	9840 Ft.	
				9800 Ft. TO	9900 Ft.	
12.*Wellhead Type						
13.*API Tubing		2 3/8 In. O.D.	8 Rd EUE Threads		J-55 Grade	

B. RESERVOIR DATA (FROM TEST & PRODUCTION DATA)

1.*Present Production	250 BFPD (50 BO, 200 BW)	Pumping (X)	Swabbing ()	Flowing ()
2. Bottom Hole Static Pressure	4000 PSIG AT:	9880	Ft.	
3. Bottom Hole Producing Pressure	3300 PSIG AT: 250 BFPD Calculated (not dependable)			
	PSIG AT:	BFPD		
4. GOR 440	total	S.C.F./S.T.B.	GOR through pump = 200 SCF/STB	
5.*Water Cut	80 %			
6. Oil API Gravity	37	°API	0.840 SP.GR.°	
7. Bottom Hole Temperature	190°F			
8. Water Specific Gravity	1.06			
9. Gas Specific Gravity	0.68			
10. Oil Viscosity (1)	CP.OR	40 SSU At	190 °F	
(2)	CP.OR	SSU At	°F	
11. PVT Data NONE	Solution GOR	FVF	PSIG	
	Solution GOR	FVF	PSIG	
	Solution GOR	FVF	PSIG	
	Solution GOR	FVF	PSIG	
12. Bubble Point Pressure	2000	PSIA		

C. SIZING SPECIFICATIONS

1.*Desired Production	1100 BFPD	220 BOPD	880 BWPD
2. Desired Pump (Intake) Vertical Setting Depth	9000 V.D. Ft. 9000 M.D.		
3. Desired Pump (Intake) Pressure	500 PSIG		
4. Required Wellhead Pressure	100 PSIG		
5. Gas Through Pump			
6. Primary Voltage Available	22,000 Volts	60 Hertz	
7. Casing Pressure	25 PSIG		
8. Special Problems: NONE	Sand ()	Scale ()	Corrosion () Paraffin () H ₂ S () CO ₂ ()

REMARKS: Operator desires stainless steel bands, three single phase transformers and no check no drain valve.

NOTE:

BY: _____ DATE: _____

*Indicates Data That Must Be Provided

**EXHIBIT 2.1
ELECTRIC SUBMERSIBLE PUMP
WELL DATA SHEET**

EXHIBIT 2.1 ELECTRIC SUBMERSIBLE PUMP WELL DATA SHEET

A. WELL DATA

1. Well No. & Field Name EXAMPLE PROBLEM NO. 3					Date 2/09/83
2. Well Location (County, State, Other) Viscosity Correction — Medium					
3. Installation					New (X) or Redesign ()
4. Producing Formations					
5. Formation Type (Sandstone, Limestone, Other)					
6.*API Casing	5.5 In. O.D.:	17 #/Ft.	Ft. TO	5600 Ft.	
7.*Liner	In. O.D.:	#/Ft.	Ft. TO	Ft.	
8.*Other Limiting Dimensions	In. I.D.	Ft. TO	Ft.		
9.*Open Hole		Ft. TO	Ft.		
10.*Total Depth	5600	Ft. Plugged Back Depth	5400 Ft.		
11.*Perforated Intervals		5300 Ft. TO	5400 Ft.		
		Ft. TO	Ft.		
12.*Wellhead Type & Size					
13.*API Tubing	2 3/8 In. O.D.	8 Rd. EUE	Threads	J-55	Grade

B. RESERVOIR DATA (FROM TEST & PRODUCTION DATA)

1.*Present Production	BOPD	BWPD	300 BFPD	Pumping (X)	Flowing ()
2. Bottom Hole Static Pressure	1700 PSIG AT:		5350 Ft.		
3. Bottom Hole Producing Pressure	1200 PSIG AT:		300 BFPD		
	PSIG AT:		BFPD		
4. GOR	50		S.C.F./S.T.B.		
5.*Water Cut	20.0%				
6. Oil API Gravity	15.0 °API				
7. Water Specific Gravity	1.06				
8. Gas Specific Gravity					
9. Bottom Hole Temperature	140°F				
10. Oil Viscosity (1) Not Avail.	CP.OR		SSU At	195 °F	
(2)	CP.OR		SSU At	°F	
11. PVT Data	Solution GOR		FVF	PSIG	
	Solution GOR		FVF	PSIG	
	Solution GOR		FVF	PSIG	
	Solution GOR		FVF	PSIG	
12. Bubble Point Pressure	PSIG				

C. SIZING SPECIFICATIONS

1.*Desired Production	800 STB	BFPD	640 BOPD	160 BWPD
2. Desired Pump (Intake) Vertical Setting Depth	5200 Ft.			
3. Desired Pump (Intake) Pressure	300 PSIG			
4. Required Wellhead Pressure	50 PSIG			
5. Gas Through Pump	% (Of Total Gas)			
6. Primary Voltage Available	14,400/24,940	Volts	60	Hertz
7. Casing Pressure	PSIG			
8. Special Problems:	Sand ()	Scale ()	Corrosion ()	Paraffin () H ₂ S () CO ₂ ()

REMARKS: Operator desires monel bands, coated equipment, three single phase transformers.
No check or drain valve.

BY: _____ DATE: _____

*Indicates Data That Must Be Provided

**EXHIBIT 3.1
ELECTRIC SUBMERSIBLE PUMP
WELL DATA SHEET**

EXHIBIT 3.1 ELECTRIC SUBMERSIBLE PUMP WELL DATA SHEET